BAN502_Project

Phase1_VariableSelection

Load libraries

```
library(tidyverse)
```

```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.
0.0 ──
## ✓ dplyr     1.2.1     ✓ readr     2.2.0
## ✓ forcats   1.0.1     ✓ stringr   1.6.0
## ✓ ggplot2   4.0.3     ✓ tibble    3.3.1
## ✓ lubridate 1.9.5     ✓ tidyr     1.3.2
## ✓ purrr     1.2.2
## ── Conflicts ────────────────────────────────────────── tidyverse_conflict
s() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
 conflicts to become errors
```

```
library(tidymodels)
```

```
## ── Attaching packages ─────────────────────────────────── tidymodels 1.
5.0 ──
## ✓ broom        1.0.12     ✓ rsample      1.3.2
## ✓ dials        1.4.3      ✓ tailor       0.1.0
## ✓ infer        1.1.0      ✓ tune         2.1.0
## ✓ modeldata    1.5.1      ✓ workflows    1.3.0
## ✓ parsnip      1.6.0      ✓ workflowsets 1.1.1
## ✓ recipes      1.3.2      ✓ yardstick    1.4.0
## ── Conflicts ────────────────────────────────────── tidymodels_conflict
s() ──
## ✖ scales::discard() masks purrr::discard()
## ✖ dplyr::filter()   masks stats::filter()
## ✖ recipes::fixed()  masks stringr::fixed()
## ✖ dplyr::lag()      masks stats::lag()
## ✖ yardstick::spec() masks readr::spec()
## ✖ recipes::step()   masks stats::step()
```

```
library(GGally)
library(ggcorrplot)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(lmtest)
```

```
## Loading required package: zoo
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(glmnet)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## Loaded glmnet 5.0
```

```
library(MASS)
```

```
##
## Attaching package: 'MASS'
##
## The following object is masked from 'package:dplyr':
##
##     select
```

```
library(leaps)
library(splines)
library(car)
```

```
## Loading required package: carData
##
## Attaching package: 'car'
##
## The following object is masked from 'package:dplyr':
##
##     recode
##
## The following object is masked from 'package:purrr':
```

```
##
##     some
```

```
library(e1071)
```

```
##
## Attaching package: 'e1071'
##
## The following object is masked from 'package:tune':
##
##     tune
##
## The following object is masked from 'package:rsample':
##
##     permutations
##
## The following object is masked from 'package:parsnip':
##
##     tune
##
## The following object is masked from 'package:ggplot2':
##
##     element
```

```
library(ROCR)
library(usemodels)
library(mice)
```

```
## Registered S3 method overwritten by 'lme4':
##   method           from
##   na.action.merMod car
##
## Attaching package: 'mice'
##
## The following object is masked from 'package:stats':
##
##     filter
##
## The following objects are masked from 'package:base':
##
##     cbind, rbind
```

```
library(VIM)
```

```
## Loading required package: colorspace
## Loading required package: grid
## VIM is ready to use.
##
## Suggestions and bug-reports can be submitted at: https://github.com/statis
tikat/VIM/issues
##
```

```
## Attaching package: 'VIM'
##
## The following object is masked from 'package:recipes':
##
##     prepare
##
## The following object is masked from 'package:datasets':
##
##     sleep
```

```
library(skimr)
library(UpSetR)
library(rpart)
```

```
##
## Attaching package: 'rpart'
##
## The following object is masked from 'package:dials':
##
##     prune
```

```
library(rpart.plot)
library(RColorBrewer)
library(rattle)
```

```
## Loading required package: bitops
##
## Attaching package: 'bitops'
##
## The following object is masked from 'package:Matrix':
##
##     %&%
##
## Rattle: A free graphical interface for data science with R.
## Version 5.6.2 Copyright (c) 2006-2023 Togaware Pty Ltd.
## Type 'rattle()' to shake, rattle, and roll your data.
##
## Attaching package: 'rattle'
##
## The following object is masked from 'package:VIM':
##
##     wine
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'lattice'
##
## The following object is masked from 'package:UpSetR':
##
```

```
##     histogram
##
##
## Attaching package: 'caret'
##
## The following objects are masked from 'package:yardstick':
##
##     precision, recall, sensitivity, specificity
##
## The following object is masked from 'package:rsample':
##
##     calibration
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(ranger)
```

```
##
## Attaching package: 'ranger'
##
## The following object is masked from 'package:rattle':
##
##     importance
```

```
library(randomForest)
```

```
## randomForest 4.7-1.2
## Type rfNews() to see new features/changes/bug fixes.
##
## Attaching package: 'randomForest'
##
## The following object is masked from 'package:ranger':
##
##     importance
##
## The following object is masked from 'package:rattle':
##
##     importance
##
## The following object is masked from 'package:gridExtra':
##
##     combine
##
## The following object is masked from 'package:dplyr':
##
##     combine
##
## The following object is masked from 'package:ggplot2':
```

```
##
##     margin
```

```
library(vip)
```

```
##
## Attaching package: 'vip'
##
## The following object is masked from 'package:utils':
##
##     vi
```

```
library(titanic)
library(naniar)
```

```
##
## Attaching package: 'naniar'
##
## The following object is masked from 'package:skimr':
##
##     n_complete
```

Import dataset

```
train <- read_csv("C:/Users/miche/OneDrive/Desktop/BAN 502 Predictive Analyti
cs/Course Project/train.csv")
```

```
## Rows: 26570 Columns: 26
## ── Column specification ────────────────────────────────────────────────────
──────
## Delimiter: ","
## chr  (4): product_code, attribute_0, attribute_1, failure
## dbl (22): id, loading, attribute_2, attribute_3, measurement_0, measuremen
t_...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this m
essage.
```

Dataset train, structure and summary

```
str(train)
```

```
## spc_tbl_ [26,570 × 26] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
##  $ id            : num [1:26570] 0 1 2 3 4 5 6 7 8 9 ...
##  $ product_code  : chr [1:26570] "A" "A" "A" "A" ...
##  $ loading       : num [1:26570] 80.1 84.9 82.4 101.1 188.1 ...
##  $ attribute_0   : chr [1:26570] "material_7" "material_7" "material_7" "m
aterial_7" ...
##  $ attribute_1   : chr [1:26570] "material_8" "material_8" "material_8" "m
aterial_8" ...
##  $ attribute_2   : num [1:26570] 9 9 9 9 9 9 9 9 9 9 ...
```

```
##  $ attribute_3   : num [1:26570] 5 5 5 5 5 5 5 5 5 5 ...
##  $ measurement_0 : num [1:26570] 7 14 12 13 9 11 12 4 9 10 ...
##  $ measurement_1 : num [1:26570] 8 3 1 2 2 4 2 8 6 4 ...
##  $ measurement_2 : num [1:26570] 4 3 5 6 8 0 4 8 5 7 ...
##  $ measurement_3 : num [1:26570] 18 18.2 18.1 17.3 19.3 ...
##  $ measurement_4 : num [1:26570] 12.5 11.5 11.7 11.2 12.9 ...
##  $ measurement_5 : num [1:26570] 15.7 17.7 16.7 18.6 17 ...
##  $ measurement_6 : num [1:26570] 19.3 17.9 18.2 18.3 15.7 ...
##  $ measurement_7 : num [1:26570] 11.7 12.7 12.7 12.6 11.3 ...
##  $ measurement_8 : num [1:26570] 20.2 17.9 18.3 19.1 18.1 ...
##  $ measurement_9 : num [1:26570] 10.7 12.4 12.7 12.5 10.3 ...
##  $ measurement_10: num [1:26570] 15.9 17.9 15.6 16.3 17.1 ...
##  $ measurement_11: num [1:26570] 17.6 17.9 NA 18.4 19.9 ...
##  $ measurement_12: num [1:26570] 15.2 11.8 13.8 10 12.4 ...
##  $ measurement_13: num [1:26570] 15 14.7 16.7 15.2 16.2 ...
##  $ measurement_14: num [1:26570] NA 15.4 18.6 15.6 12.8 ...
##  $ measurement_15: num [1:26570] 13 14.4 14.1 16.2 13.2 ...
##  $ measurement_16: num [1:26570] 14.7 15.6 17.9 17.2 16.4 ...
##  $ measurement_17: num [1:26570] 764 682 663 826 580 ...
##  $ failure       : chr [1:26570] "No" "No" "No" "No" ...
##  - attr(*, "spec")=
##   .. cols(
##   ..   id = col_double(),
##   ..   product_code = col_character(),
##   ..   loading = col_double(),
##   ..   attribute_0 = col_character(),
##   ..   attribute_1 = col_character(),
##   ..   attribute_2 = col_double(),
##   ..   attribute_3 = col_double(),
##   ..   measurement_0 = col_double(),
##   ..   measurement_1 = col_double(),
##   ..   measurement_2 = col_double(),
##   ..   measurement_3 = col_double(),
##   ..   measurement_4 = col_double(),
##   ..   measurement_5 = col_double(),
##   ..   measurement_6 = col_double(),
##   ..   measurement_7 = col_double(),
##   ..   measurement_8 = col_double(),
##   ..   measurement_9 = col_double(),
##   ..   measurement_10 = col_double(),
##   ..   measurement_11 = col_double(),
##   ..   measurement_12 = col_double(),
##   ..   measurement_13 = col_double(),
##   ..   measurement_14 = col_double(),
##   ..   measurement_15 = col_double(),
##   ..   measurement_16 = col_double(),
##   ..   measurement_17 = col_double(),
##   ..   failure = col_character()
##   .. )
##  - attr(*, "problems")=<pointer: 0x00000257d8be74b0>
```

```
summary(train)
```

```
##        id           product_code      loading          attribute_0
##  Min.   :    0   Length   :26570   Min.   : 33.16   Length   :26570
##  1st Qu.: 6642   N.unique :    5   1st Qu.: 99.99   N.unique :    2
##  Median :13284   N.blank  :    0   Median :122.39   N.blank  :    0
##  Mean   :13284   Min.nchar:    1   Mean   :127.83   Min.nchar:   10
##  3rd Qu.:19927   Max.nchar:    1   3rd Qu.:149.15   Max.nchar:   10
##  Max.   :26569                     Max.   :385.86
##                                    NAs    :250
##     attribute_1     attribute_2     attribute_3   measurement_0
##  Length   :26570   Min.   :5.000   Min.   :5.00   Min.   : 0.000
##  N.unique :    3   1st Qu.:6.000   1st Qu.:6.00   1st Qu.: 4.000
##  N.blank  :    0   Median :6.000   Median :8.00   Median : 7.000
##  Min.nchar:   10   Mean   :6.754   Mean   :7.24   Mean   : 7.416
##  Max.nchar:   10   3rd Qu.:8.000   3rd Qu.:8.00   3rd Qu.:10.000
##                    Max.   :9.000   Max.   :9.00   Max.   :29.000
##
##  measurement_1    measurement_2    measurement_3   measurement_4
##  Min.   : 0.000   Min.   : 0.000   Min.   :13.97   Min.   : 8.008
##  1st Qu.: 5.000   1st Qu.: 4.000   1st Qu.:17.12   1st Qu.:11.051
##  Median : 8.000   Median : 6.000   Median :17.79   Median :11.733
##  Mean   : 8.233   Mean   : 6.257   Mean   :17.79   Mean   :11.732
##  3rd Qu.:11.000   3rd Qu.: 8.000   3rd Qu.:18.47   3rd Qu.:12.410
##  Max.   :29.000   Max.   :24.000   Max.   :21.50   Max.   :16.484
##                                    NAs    :381     NAs    :538
##  measurement_5   measurement_6   measurement_7    measurement_8
##  Min.   :12.07   Min.   :12.71   Min.   : 7.968   Min.   :15.22
##  1st Qu.:16.44   1st Qu.:16.84   1st Qu.:11.045   1st Qu.:18.34
##  Median :17.13   Median :17.52   Median :11.712   Median :19.02
##  Mean   :17.13   Mean   :17.51   Mean   :11.717   Mean   :19.02
##  3rd Qu.:17.80   3rd Qu.:18.18   3rd Qu.:12.391   3rd Qu.:19.71
##  Max.   :21.43   Max.   :21.54   Max.   :15.419   Max.   :23.81
##  NAs    :676     NAs    :796     NAs    :937     NAs    :1048
##  measurement_9    measurement_10   measurement_11  measurement_12
##  Min.   : 7.537   Min.   : 9.323   Min.   :12.46   Min.   : 5.167
##  1st Qu.:10.757   1st Qu.:15.209   1st Qu.:18.17   1st Qu.:10.703
##  Median :11.430   Median :16.127   Median :19.21   Median :11.717
##  Mean   :11.431   Mean   :16.118   Mean   :19.17   Mean   :11.702
##  3rd Qu.:12.102   3rd Qu.:17.025   3rd Qu.:20.21   3rd Qu.:12.709
##  Max.   :15.412   Max.   :22.479   Max.   :25.64   Max.   :17.663
##  NAs    :1227     NAs    :1300     NAs    :1468    NAs    :1601
##  measurement_13  measurement_14  measurement_15   measurement_16
##  Min.   :10.89   Min.   : 9.14   Min.   : 9.104   Min.   : 9.701
##  1st Qu.:14.89   1st Qu.:15.06   1st Qu.:13.957   1st Qu.:15.268
##  Median :15.63   Median :16.04   Median :14.969   Median :16.436
##  Mean   :15.65   Mean   :16.05   Mean   :14.996   Mean   :16.461
##  3rd Qu.:16.37   3rd Qu.:17.08   3rd Qu.:16.018   3rd Qu.:17.628
##  Max.   :22.71   Max.   :22.30   Max.   :21.626   Max.   :24.094
##  NAs    :1774    NAs    :1874    NAs    :2009     NAs    :2110
```

```
##  measurement_17        failure
##  Min.   : 196.8   Length   :26570
##  1st Qu.: 619.0   N.unique :    2
##  Median : 701.0   N.blank  :    0
##  Mean   : 701.3   Min.nchar:    2
##  3rd Qu.: 784.1   Max.nchar:    3
##  Max.   :1312.8
##  NAs    :2284
```

Dataset train, factor conversion

```
train = train %>% mutate(product_code = as_factor(product_code)) %>%
  mutate(attribute_0 = as_factor(attribute_0)) %>%
  mutate(attribute_1 = as_factor(attribute_1)) %>%
  mutate(failure = as_factor(failure))
```

Dataset train, structure and summary

```
str(train)
```

```
## tibble [26,570 × 26] (S3: tbl_df/tbl/data.frame)
##  $ id            : num [1:26570] 0 1 2 3 4 5 6 7 8 9 ...
##  $ product_code  : Factor w/ 5 levels "A","B","C","D",..: 1 1 1 1 1 1 1 1
1 1 ...
##  $ loading       : num [1:26570] 80.1 84.9 82.4 101.1 188.1 ...
##  $ attribute_0   : Factor w/ 2 levels "material_7","material_5": 1 1 1 1 1
 1 1 1 1 1 ...
##  $ attribute_1   : Factor w/ 3 levels "material_8","material_5",..: 1 1 1
1 1 1 1 1 1 1 ...
##  $ attribute_2   : num [1:26570] 9 9 9 9 9 9 9 9 9 9 ...
##  $ attribute_3   : num [1:26570] 5 5 5 5 5 5 5 5 5 5 ...
##  $ measurement_0 : num [1:26570] 7 14 12 13 9 11 12 4 9 10 ...
##  $ measurement_1 : num [1:26570] 8 3 1 2 2 4 2 8 6 4 ...
##  $ measurement_2 : num [1:26570] 4 3 5 6 8 0 4 8 5 7 ...
##  $ measurement_3 : num [1:26570] 18 18.2 18.1 17.3 19.3 ...
##  $ measurement_4 : num [1:26570] 12.5 11.5 11.7 11.2 12.9 ...
##  $ measurement_5 : num [1:26570] 15.7 17.7 16.7 18.6 17 ...
##  $ measurement_6 : num [1:26570] 19.3 17.9 18.2 18.3 15.7 ...
##  $ measurement_7 : num [1:26570] 11.7 12.7 12.7 12.6 11.3 ...
##  $ measurement_8 : num [1:26570] 20.2 17.9 18.3 19.1 18.1 ...
##  $ measurement_9 : num [1:26570] 10.7 12.4 12.7 12.5 10.3 ...
##  $ measurement_10: num [1:26570] 15.9 17.9 15.6 16.3 17.1 ...
##  $ measurement_11: num [1:26570] 17.6 17.9 NA 18.4 19.9 ...
##  $ measurement_12: num [1:26570] 15.2 11.8 13.8 10 12.4 ...
##  $ measurement_13: num [1:26570] 15 14.7 16.7 15.2 16.2 ...
##  $ measurement_14: num [1:26570] NA 15.4 18.6 15.6 12.8 ...
##  $ measurement_15: num [1:26570] 13 14.4 14.1 16.2 13.2 ...
##  $ measurement_16: num [1:26570] 14.7 15.6 17.9 17.2 16.4 ...
##  $ measurement_17: num [1:26570] 764 682 663 826 580 ...
##  $ failure       : Factor w/ 2 levels "No","Yes": 1 1 1 1 1 1 2 2 1 1 ...
```

```
summary(train)
```

```
##        id         product_code    loading           attribute_0
##  Min.   :    0   A:5100       Min.   : 33.16   material_7:21320
##  1st Qu.: 6642   B:5250       1st Qu.: 99.99   material_5: 5250
##  Median :13284   C:5765       Median :122.39
##  Mean   :13284   D:5112       Mean   :127.83
##  3rd Qu.:19927   E:5343       3rd Qu.:149.15
##  Max.   :26569                Max.   :385.86
##                               NAs    :250
##      attribute_1     attribute_2     attribute_3   measurement_0
##  material_8:10865   Min.   :5.000   Min.   :5.00   Min.   : 0.000
##  material_5:10362   1st Qu.:6.000   1st Qu.:6.00   1st Qu.: 4.000
##  material_6: 5343   Median :6.000   Median :8.00   Median : 7.000
##                     Mean   :6.754   Mean   :7.24   Mean   : 7.416
##                     3rd Qu.:8.000   3rd Qu.:8.00   3rd Qu.:10.000
##                     Max.   :9.000   Max.   :9.00   Max.   :29.000
##
##  measurement_1    measurement_2    measurement_3   measurement_4
##  Min.   : 0.000   Min.   : 0.000   Min.   :13.97   Min.   : 8.008
##  1st Qu.: 5.000   1st Qu.: 4.000   1st Qu.:17.12   1st Qu.:11.051
##  Median : 8.000   Median : 6.000   Median :17.79   Median :11.733
##  Mean   : 8.233   Mean   : 6.257   Mean   :17.79   Mean   :11.732
##  3rd Qu.:11.000   3rd Qu.: 8.000   3rd Qu.:18.47   3rd Qu.:12.410
##  Max.   :29.000   Max.   :24.000   Max.   :21.50   Max.   :16.484
##                                    NAs    :381     NAs    :538
##  measurement_5   measurement_6   measurement_7    measurement_8
##  Min.   :12.07   Min.   :12.71   Min.   : 7.968   Min.   :15.22
##  1st Qu.:16.44   1st Qu.:16.84   1st Qu.:11.045   1st Qu.:18.34
##  Median :17.13   Median :17.52   Median :11.712   Median :19.02
##  Mean   :17.13   Mean   :17.51   Mean   :11.717   Mean   :19.02
##  3rd Qu.:17.80   3rd Qu.:18.18   3rd Qu.:12.391   3rd Qu.:19.71
##  Max.   :21.43   Max.   :21.54   Max.   :15.419   Max.   :23.81
##  NAs    :676     NAs    :796     NAs    :937     NAs    :1048
##  measurement_9    measurement_10   measurement_11  measurement_12
##  Min.   : 7.537   Min.   : 9.323   Min.   :12.46   Min.   : 5.167
##  1st Qu.:10.757   1st Qu.:15.209   1st Qu.:18.17   1st Qu.:10.703
##  Median :11.430   Median :16.127   Median :19.21   Median :11.717
##  Mean   :11.431   Mean   :16.118   Mean   :19.17   Mean   :11.702
##  3rd Qu.:12.102   3rd Qu.:17.025   3rd Qu.:20.21   3rd Qu.:12.709
##  Max.   :15.412   Max.   :22.479   Max.   :25.64   Max.   :17.663
##  NAs    :1227     NAs    :1300     NAs    :1468    NAs    :1601
##  measurement_13  measurement_14  measurement_15   measurement_16
##  Min.   :10.89   Min.   : 9.14   Min.   : 9.104   Min.   : 9.701
##  1st Qu.:14.89   1st Qu.:15.06   1st Qu.:13.957   1st Qu.:15.268
##  Median :15.63   Median :16.04   Median :14.969   Median :16.436
##  Mean   :15.65   Mean   :16.05   Mean   :14.996   Mean   :16.461
##  3rd Qu.:16.37   3rd Qu.:17.08   3rd Qu.:16.018   3rd Qu.:17.628
##  Max.   :22.71   Max.   :22.30   Max.   :21.626   Max.   :24.094
##  NAs    :1774    NAs    :1874    NAs    :2009     NAs    :2110
```

```
##  measurement_17   failure
##  Min.   : 196.8   No :20921
##  1st Qu.: 619.0   Yes: 5649
##  Median : 701.0
##  Mean   : 701.3
##  3rd Qu.: 784.1
##  Max.   :1312.8
##  NAs    :2284
```

```
skim(train)
```

Data summary

Name	train
Number of rows	26570
Number of columns	26
_____	
Column type frequency:	
factor	4
numeric	22
_____	
Group variables	None

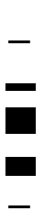
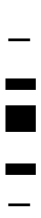
Variable type: factor

skim_variable	n_missing	complete_rate	ordered	n_unique	top_counts
product_code	0	1	FALSE	5	C: 5765, E: 5343, B: 5250, D: 5112
attribute_0	0	1	FALSE	2	mat: 21320, mat: 5250
attribute_1	0	1	FALSE	3	mat: 10865, mat: 10362, mat: 5343
failure	0	1	FALSE	2	No: 20921, Yes: 5649

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
id	0	1.00	13284.50	7670.24	0.00	6642.25	13284.50	19926.75	26569.00	▇▇▇▇▇

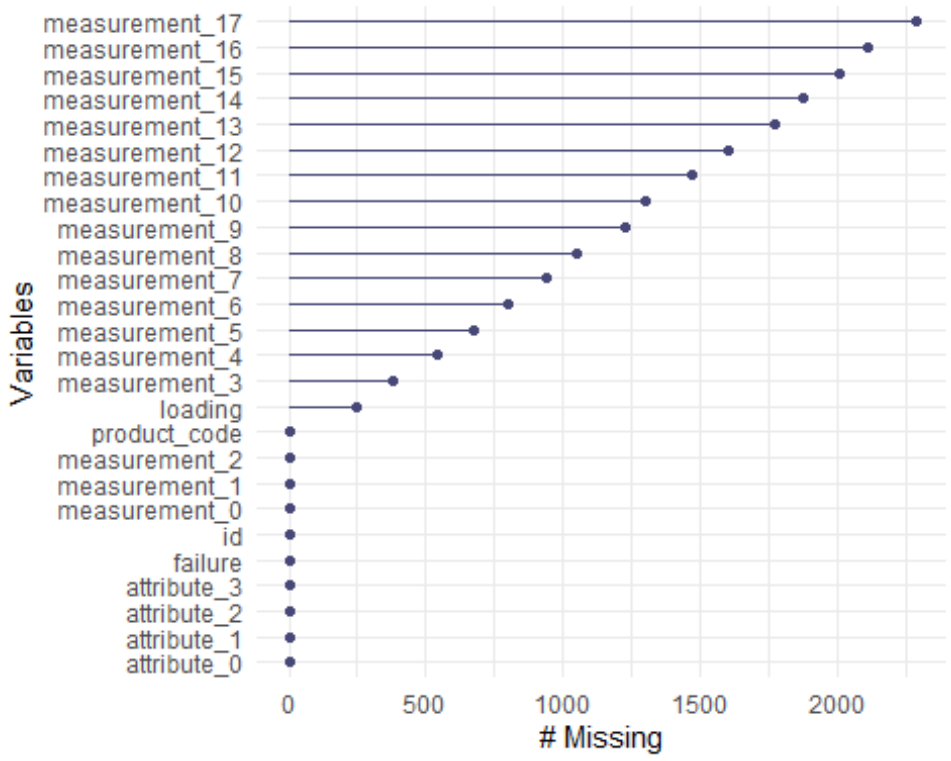
skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
loading	250	0.99	127.83	39.03	33.16	99.99	122.39	149.15	385.86	
attribute_2	0	1.00	6.75	1.47	5.00	6.00	6.00	8.00	9.00	
attribute_3	0	1.00	7.24	1.46	5.00	6.00	8.00	8.00	9.00	
measurement_0	0	1.00	7.42	4.12	0.00	4.00	7.00	10.00	29.00	
measurement_1	0	1.00	8.23	4.20	0.00	5.00	8.00	11.00	29.00	
measurement_2	0	1.00	6.26	3.31	0.00	4.00	6.00	8.00	24.00	
measurement_3	381	0.99	17.79	1.00	13.97	17.12	17.79	18.47	21.50	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_4	538	0.98	11.73	1.00	8.01	11.05	11.73	12.41	16.48	
measurement_5	676	0.97	17.13	1.00	12.07	16.44	17.13	17.80	21.42	
measurement_6	796	0.97	17.51	1.00	12.71	16.84	17.52	18.18	21.54	
measurement_7	937	0.96	11.72	1.00	7.97	11.04	11.71	12.39	15.42	
measurement_8	1048	0.96	19.02	1.01	15.22	18.34	19.02	19.71	23.81	
measurement_9	1227	0.95	11.43	1.00	7.54	10.76	11.43	12.10	15.41	
measurement_10	1300	0.95	16.12	1.41	9.32	15.21	16.13	17.02	22.48	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
measurement_11	1468	0.94	19.17	1.52	12.46	18.17	19.21	20.21	25.64	
measurement_12	1601	0.94	11.70	1.49	5.17	10.70	11.72	12.71	17.66	
measurement_13	1774	0.93	15.65	1.16	10.89	14.89	15.63	16.37	22.71	
measurement_14	1874	0.93	16.05	1.49	9.14	15.06	16.04	17.08	22.30	
measurement_15	2009	0.92	15.00	1.55	9.10	13.96	14.97	16.02	21.63	
measurement_16	2110	0.92	16.46	1.71	9.70	15.27	16.44	17.63	24.09	
measurement_17	2284	0.91	701.27	123.30	196.79	618.96	701.02	784.09	1312.79	

Explore visual of missingness

```
gg_miss_var(train)
```



Explore/ visual 1 for patterns of missingness

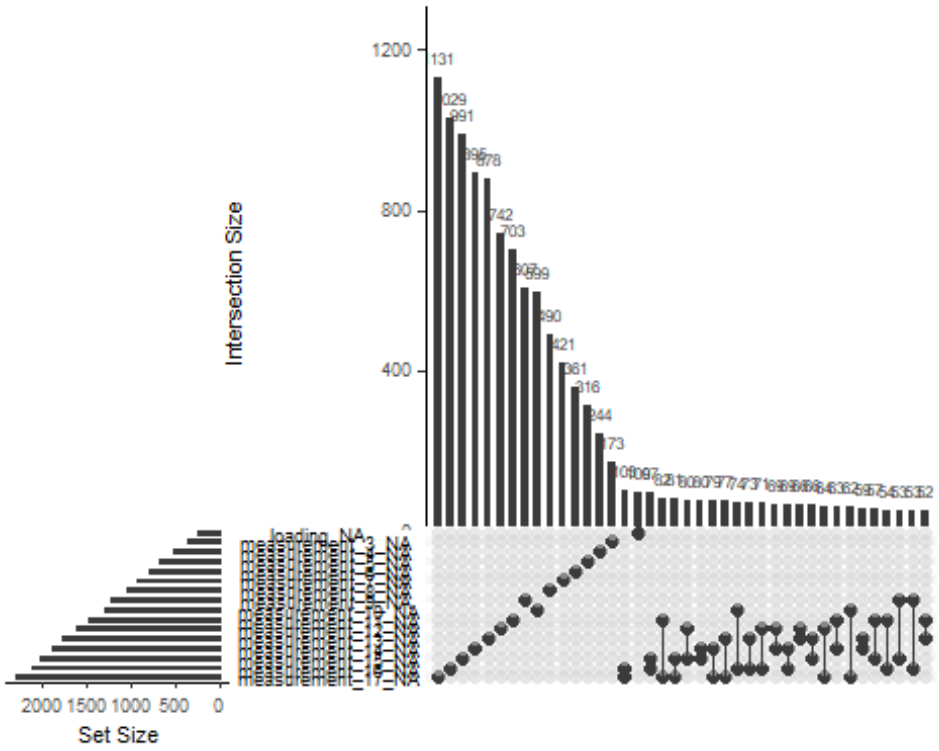
```
gg_miss_upset(train, nsets = 16)
```

```
## Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
## ℹ Please use tidy evaluation idioms with `aes()`.
## ℹ See also `vignette("ggplot2-in-packages")` for more information.
## ℹ The deprecated feature was likely used in the UpSetR package.
##   Please report the issue at <https://github.com/hms-dbmi/UpSetR/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## ℹ Please use `linewidth` instead.
## ℹ The deprecated feature was likely used in the UpSetR package.
##   Please report the issue at <https://github.com/hms-dbmi/UpSetR/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

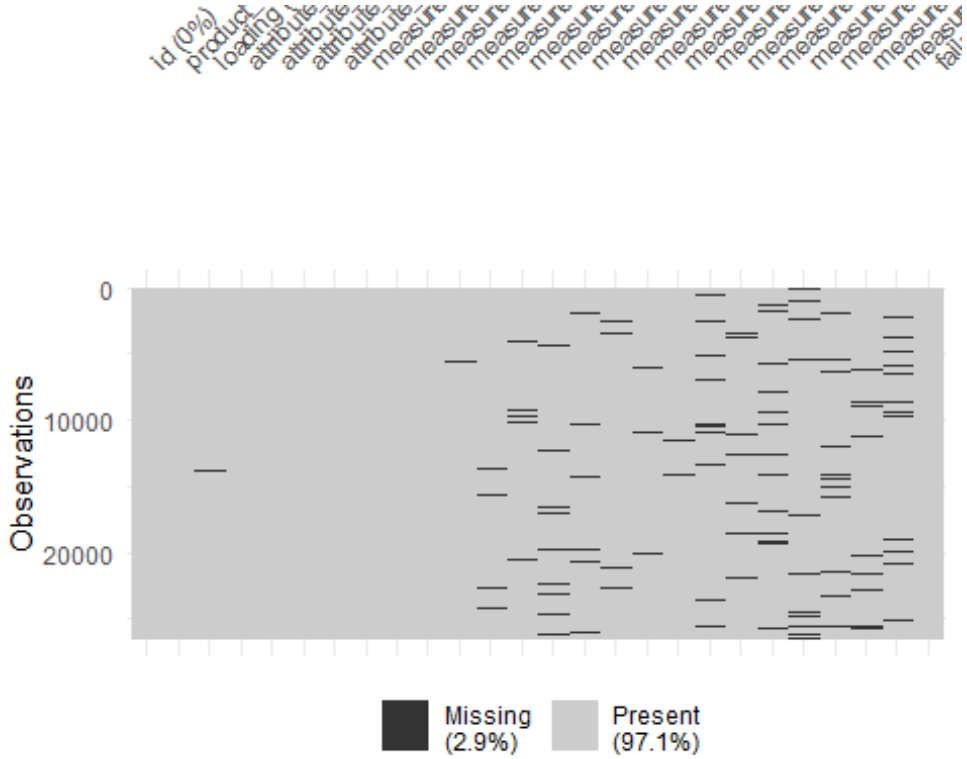
## Warning: The `size` argument of `element_line()` is deprecated as of ggplo
t2 3.4.0.
## ℹ Please use the `linewidth` argument instead.
## ℹ The deprecated feature was likely used in the UpSetR package.
##   Please report the issue at <https://github.com/hms-dbmi/UpSetR/issues>.
## This warning is displayed once per session.
```

```
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Explore/ visual 2 for patterns of missingness

```
vis_miss(train)
```

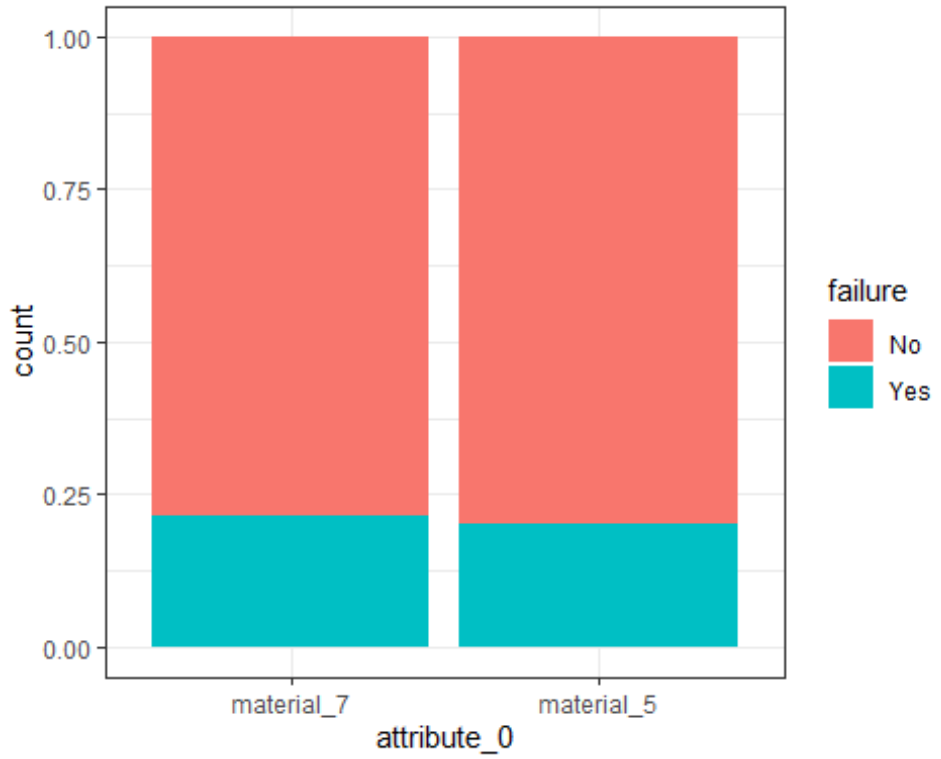


Data Exploration

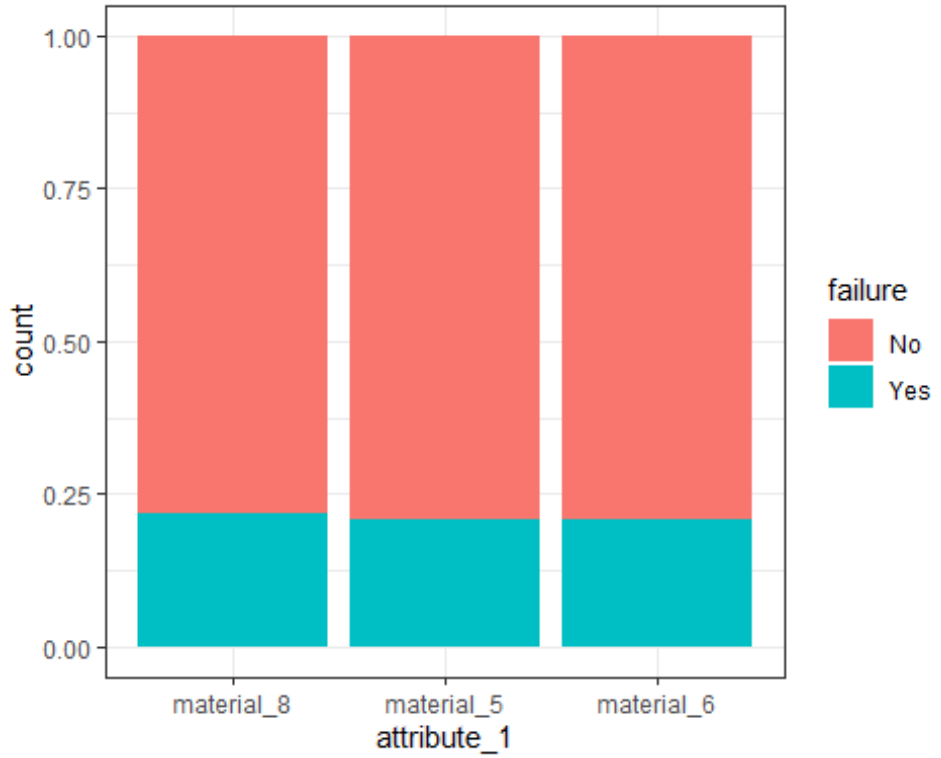
Categorical variables

Bar charts

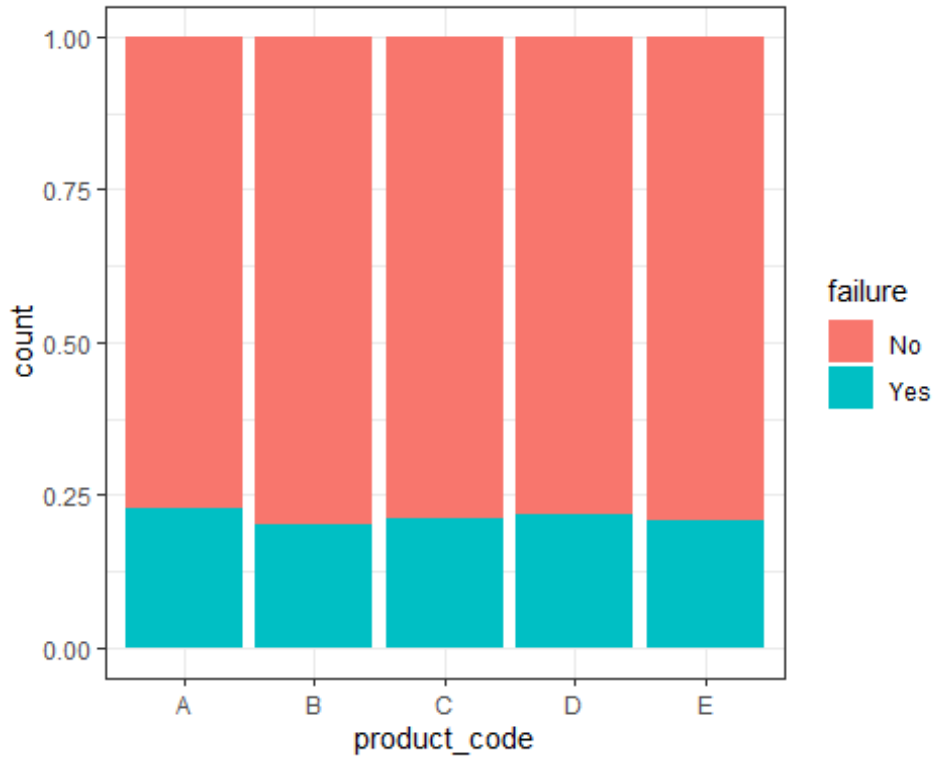
```
ggplot(train, aes(x=attribute_0, fill=failure)) + geom_bar(position="fill") +
 theme_bw()
```



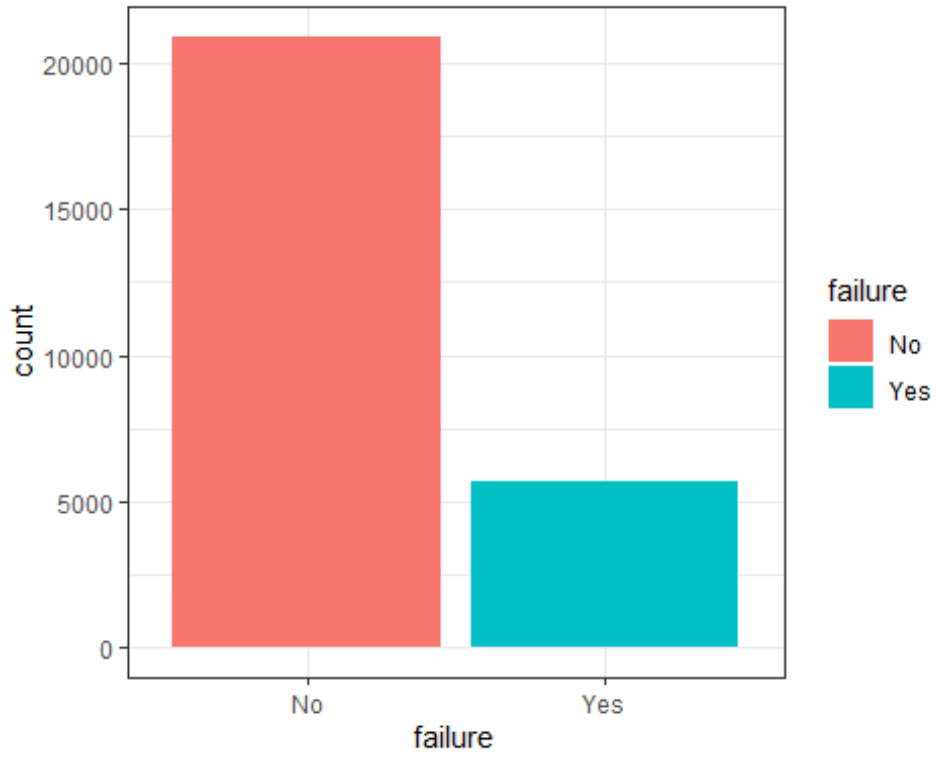

```
ggplot(train, aes(x=attribute_1, fill=failure)) + geom_bar(position = "fill")
 + theme_bw()
```



```
ggplot(train, aes(x=product_code, fill=failure)) + geom_bar(position = "fill
") + theme_bw()
```




```
ggplot(train, aes(x=failure, fill=failure)) + geom_bar() + theme_bw()
```



Proportions table, categorical variables

```
train %>%
  group_by(attribute_0) %>%
  summarise(
    n = n(),
    failures = sum(failure == "Yes"),
    failure_rate = failures / n
  )
```

```
## # A tibble: 2 × 4
##   attribute_0     n failures failure_rate
##   <fct>       <int>    <int>        <dbl>
## 1 material_7  21320     4597        0.216
## 2 material_5   5250     1052        0.200
```

```
train %>%
  group_by(attribute_1) %>%
  summarise(
    n = n(),
    failures = sum(failure == "Yes"),
    failure_rate = failures / n
  )
```

```
## # A tibble: 3 × 4
##   attribute_1     n failures failure_rate
##   <fct>       <int>    <int>        <dbl>
## 1 material_8  10865     2379        0.219
```

```
## 2 material_5  10362     2164        0.209
## 3 material_6   5343     1106        0.207
```

```
train %>%
  group_by(product_code) %>%
  summarise(
    n = n(),
    failures = sum(failure == "Yes"),
    failure_rate = failures / n
  )
```

```
## # A tibble: 5 × 4
##   product_code     n failures failure_rate
##   <fct>        <int>    <int>        <dbl>
## 1 A             5100     1159        0.227
## 2 B             5250     1052        0.200
## 3 C             5765     1220        0.212
## 4 D             5112     1112        0.218
## 5 E             5343     1106        0.207
```

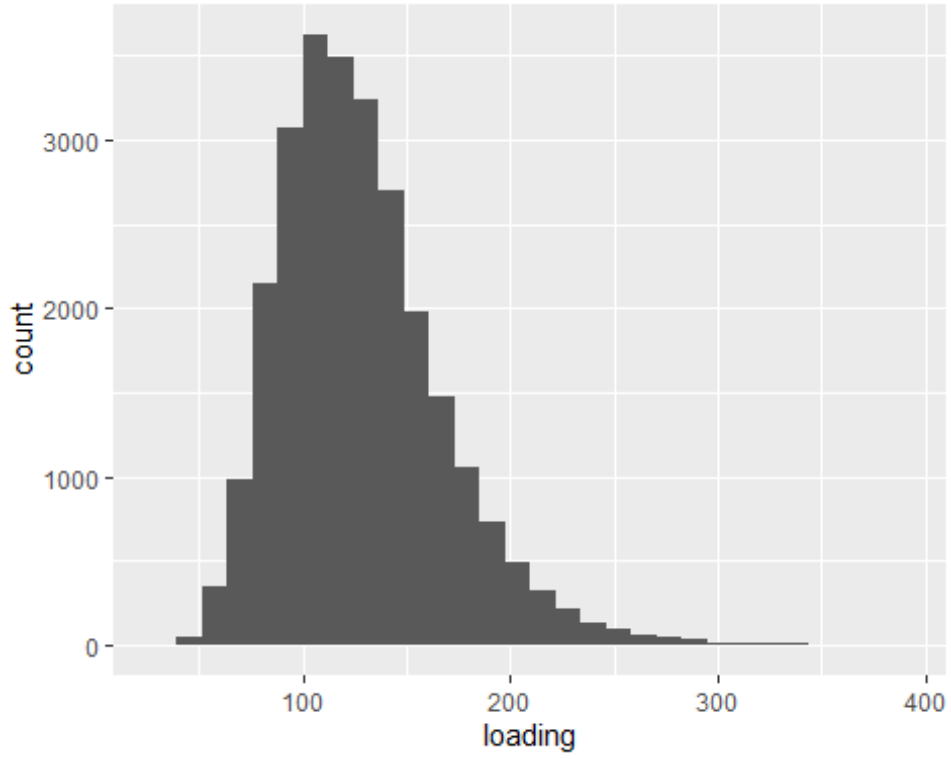
Numerical variables

Distributions

```
ggplot(train, aes(x=loading)) + geom_histogram()
```

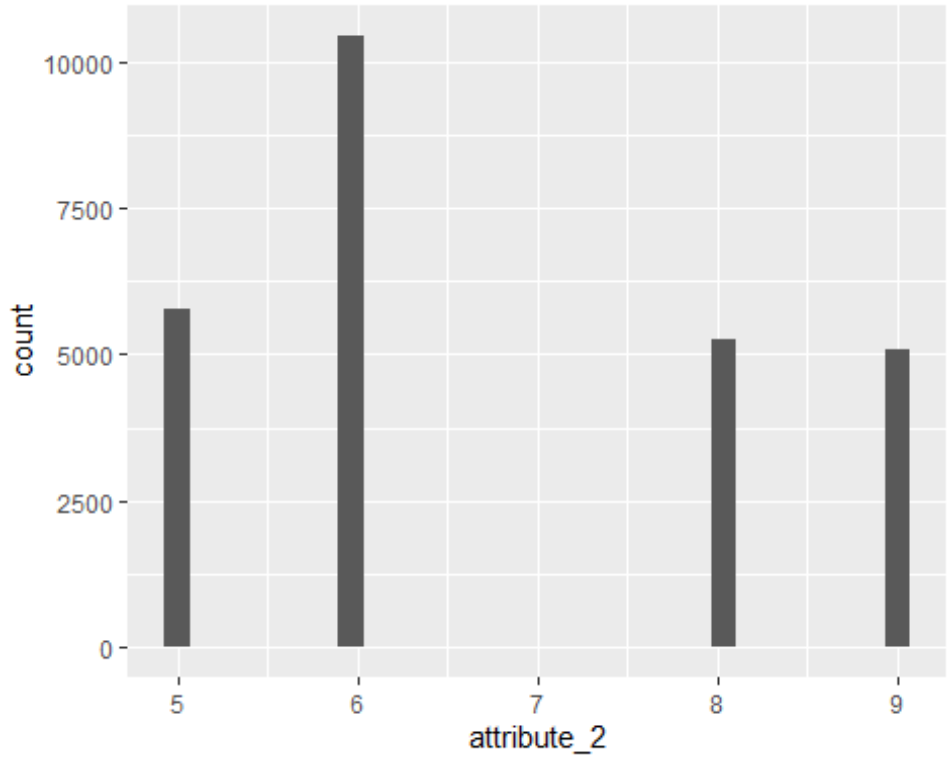
```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 250 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



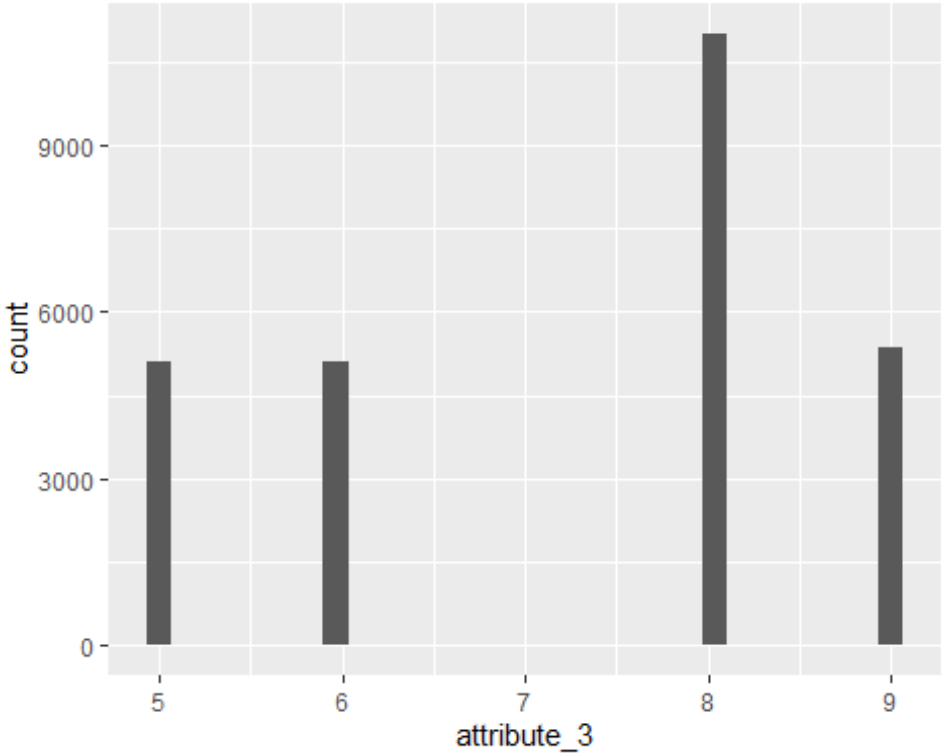

```
ggplot(train, aes(x=attribute_2)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



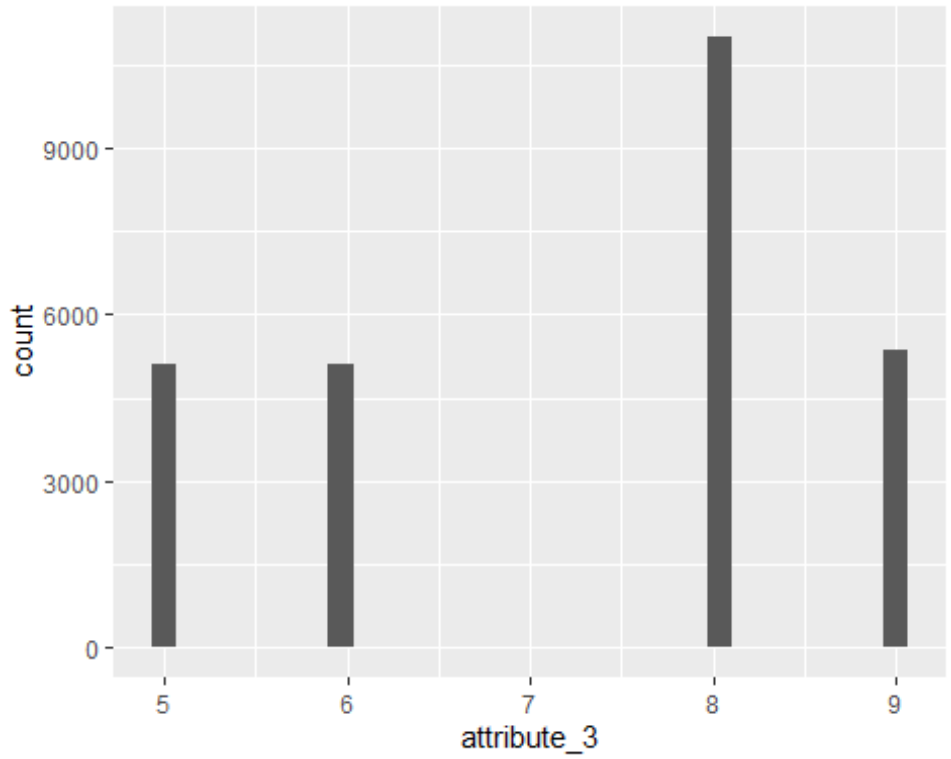
```
ggplot(train, aes(x=attribute_3)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



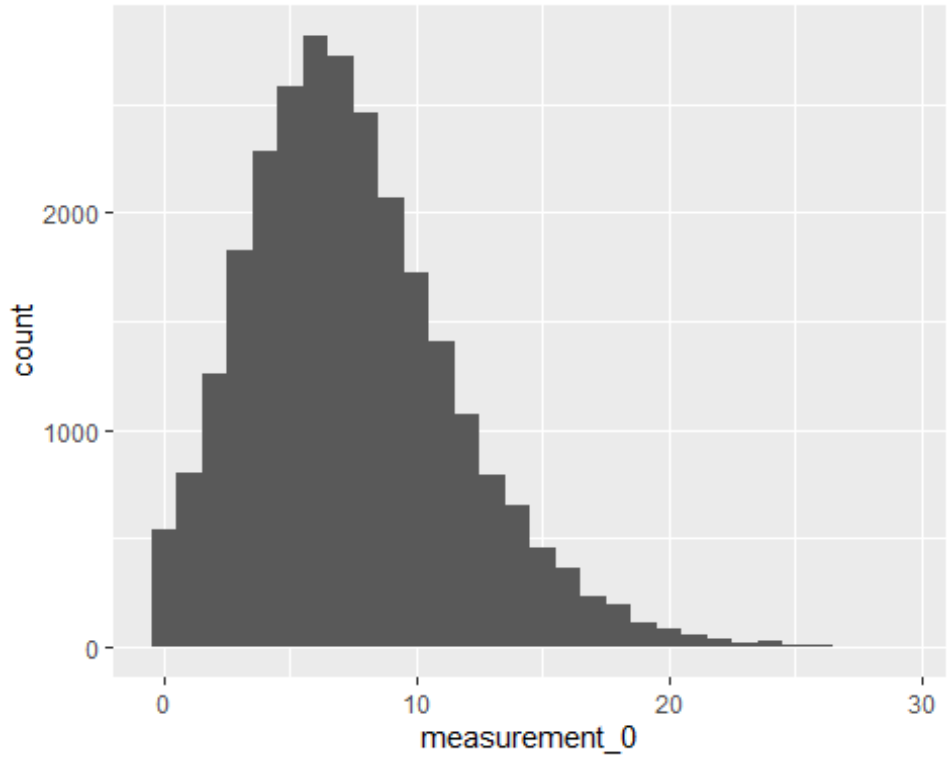

```
ggplot(train, aes(x=attribute_3)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



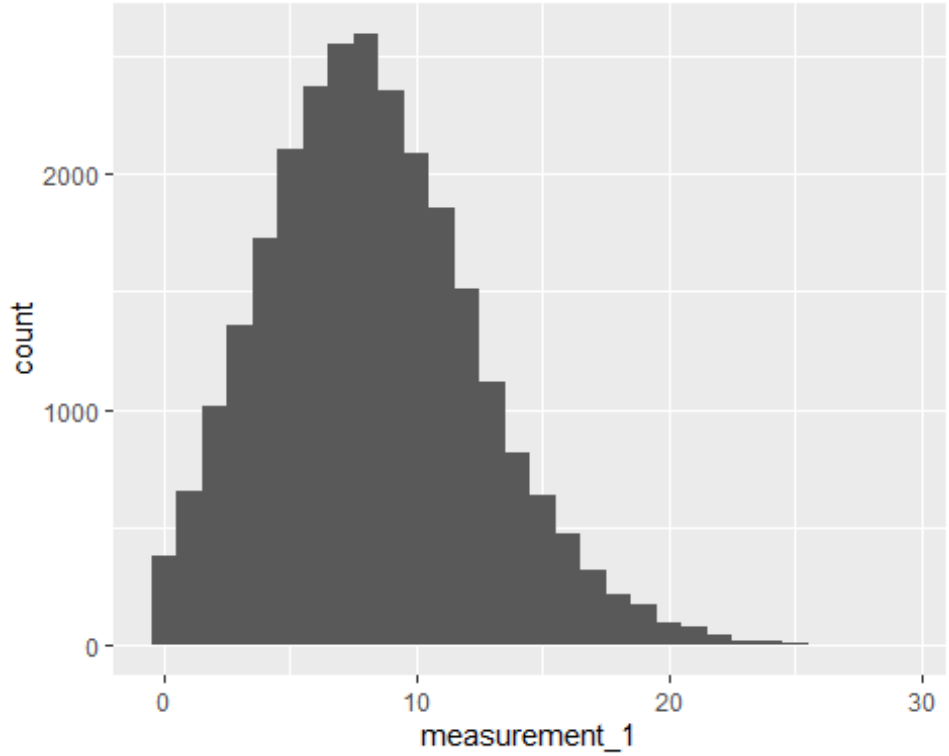

```
ggplot(train, aes(x=measurement_0)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



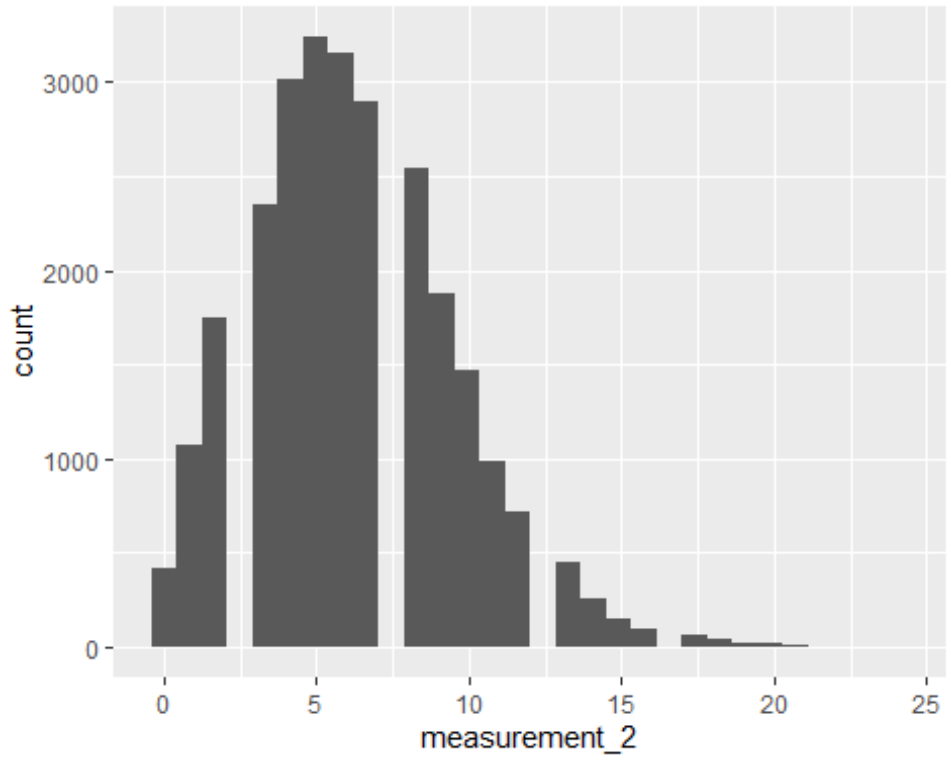
```
ggplot(train, aes(x=measurement_1)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(train, aes(x=measurement_2)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```



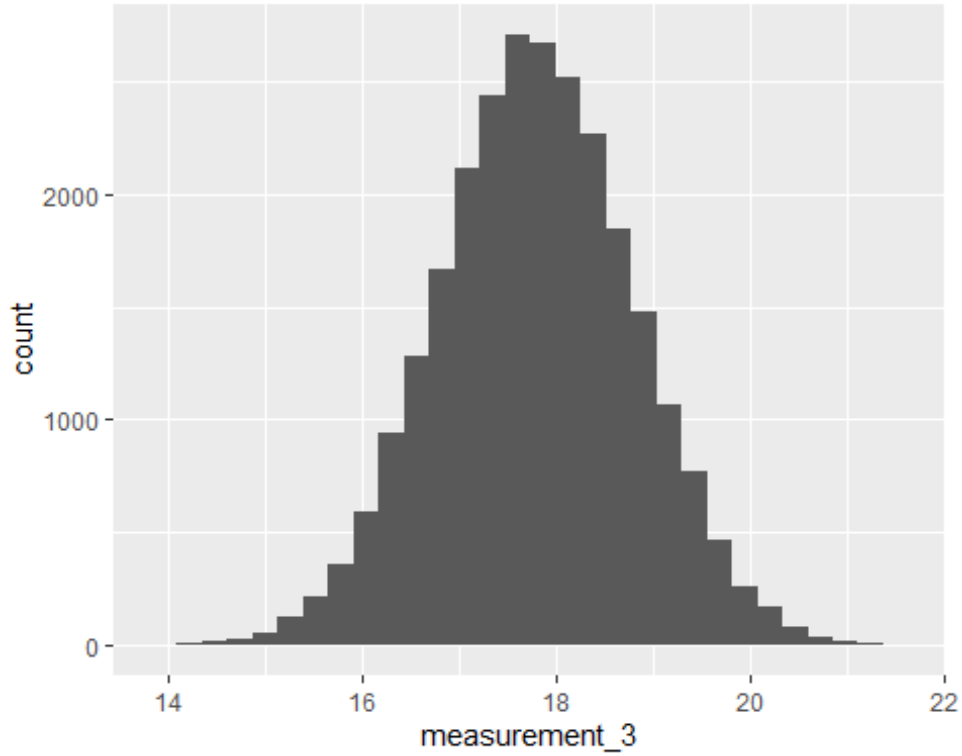
Remove outliers

```
train = train %>% filter(measurement_2 < 22)

ggplot(train, aes(x=measurement_3)) + geom_histogram()

## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.

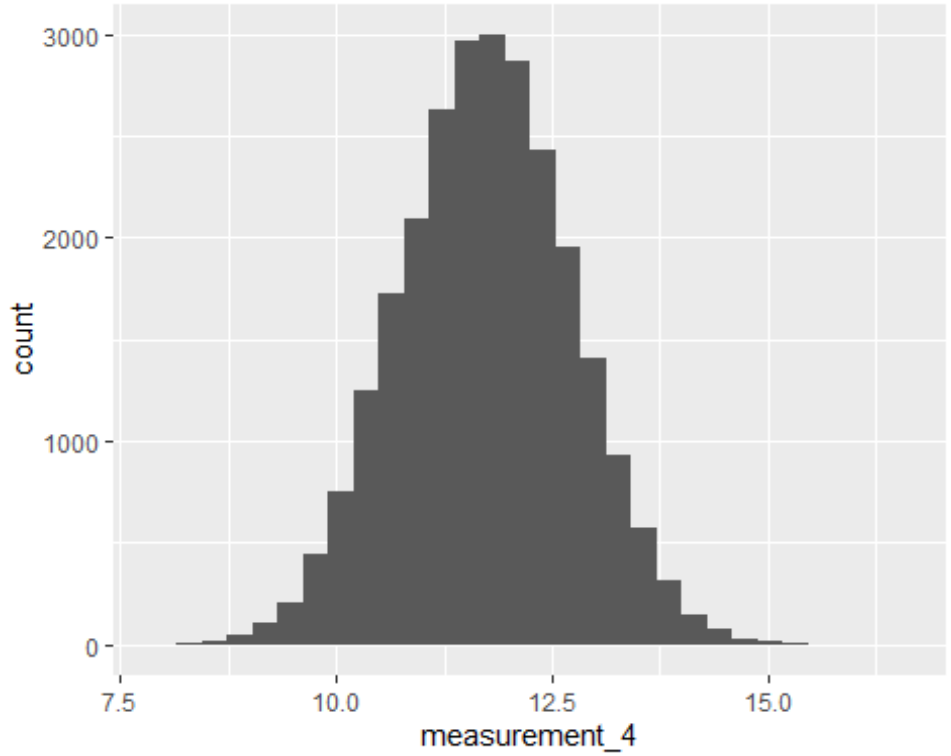
## Warning: Removed 381 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_4)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

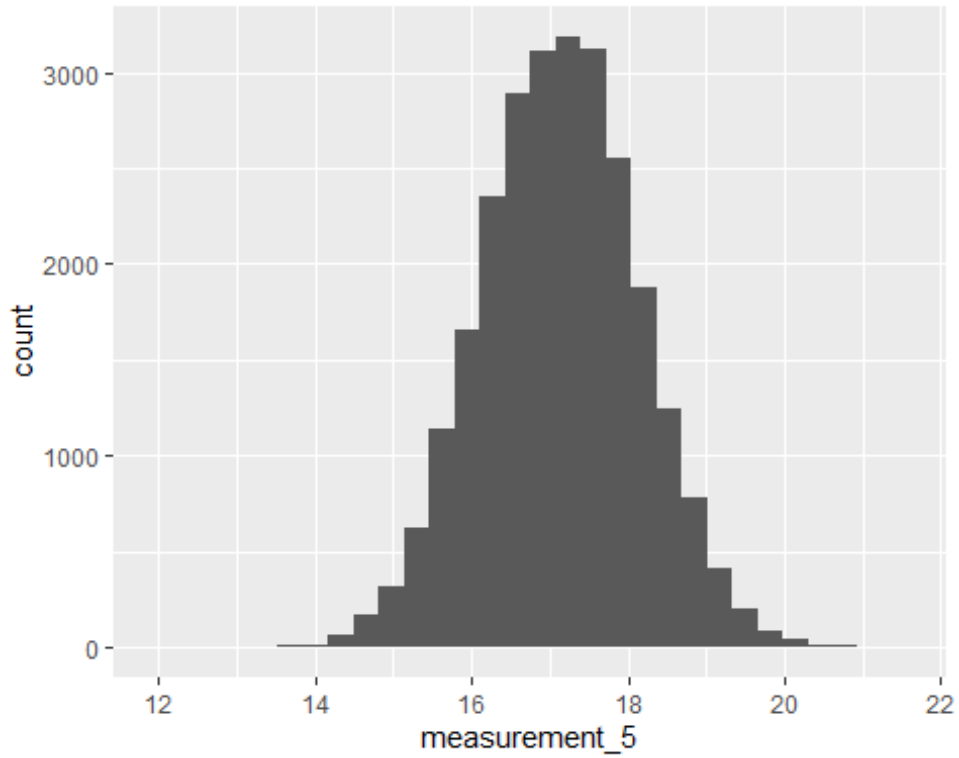
```
## Warning: Removed 538 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_5)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

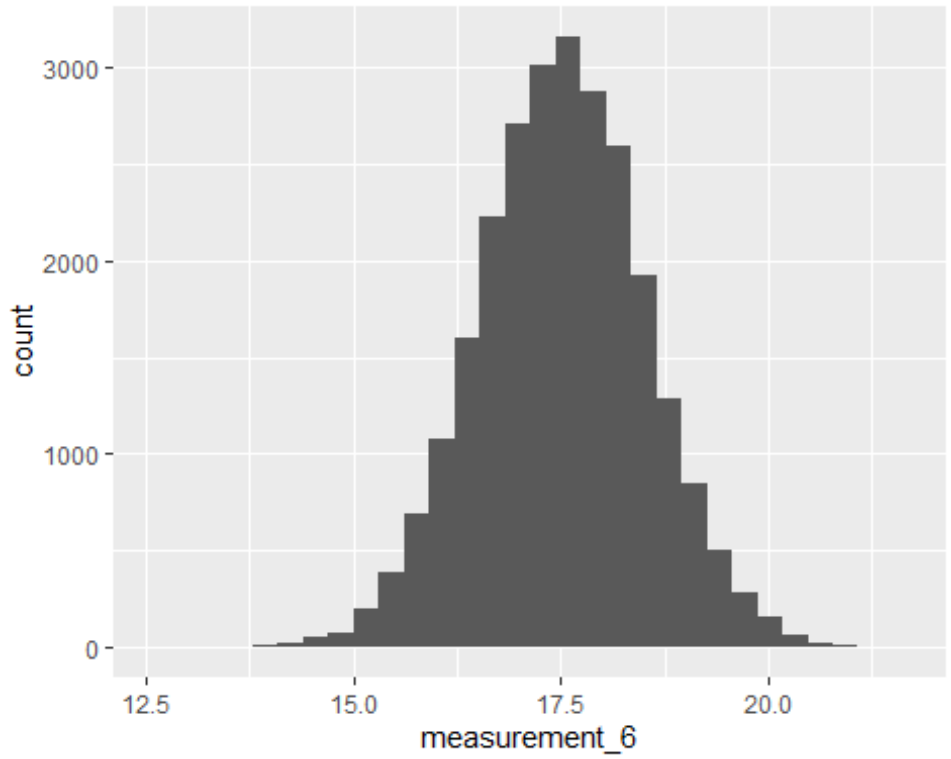
```
## Warning: Removed 676 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_6)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

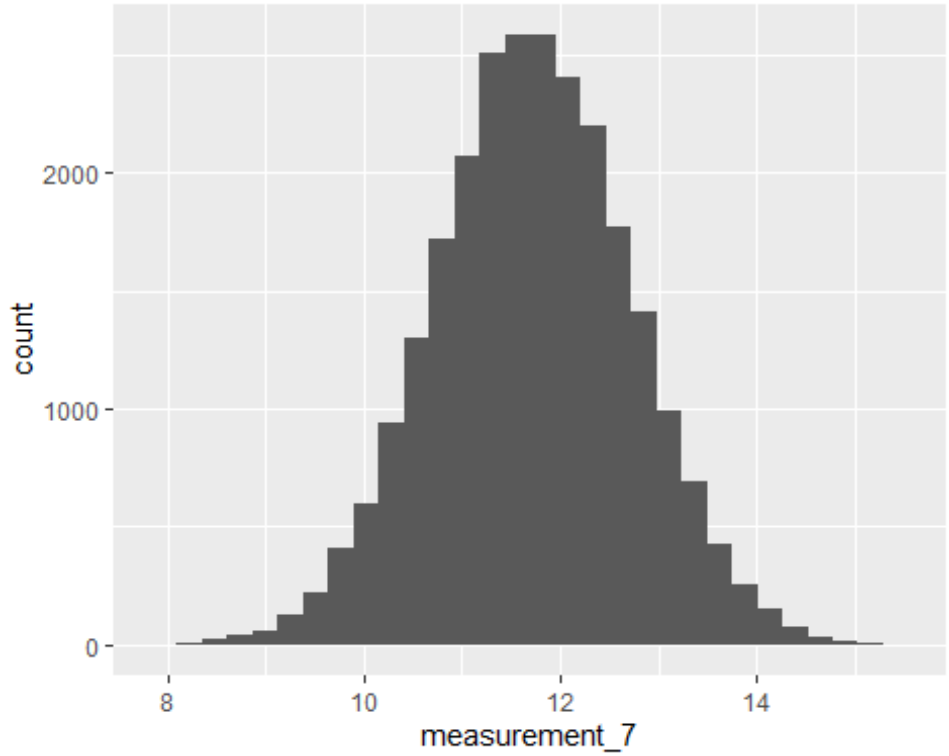
```
## Warning: Removed 796 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_7)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

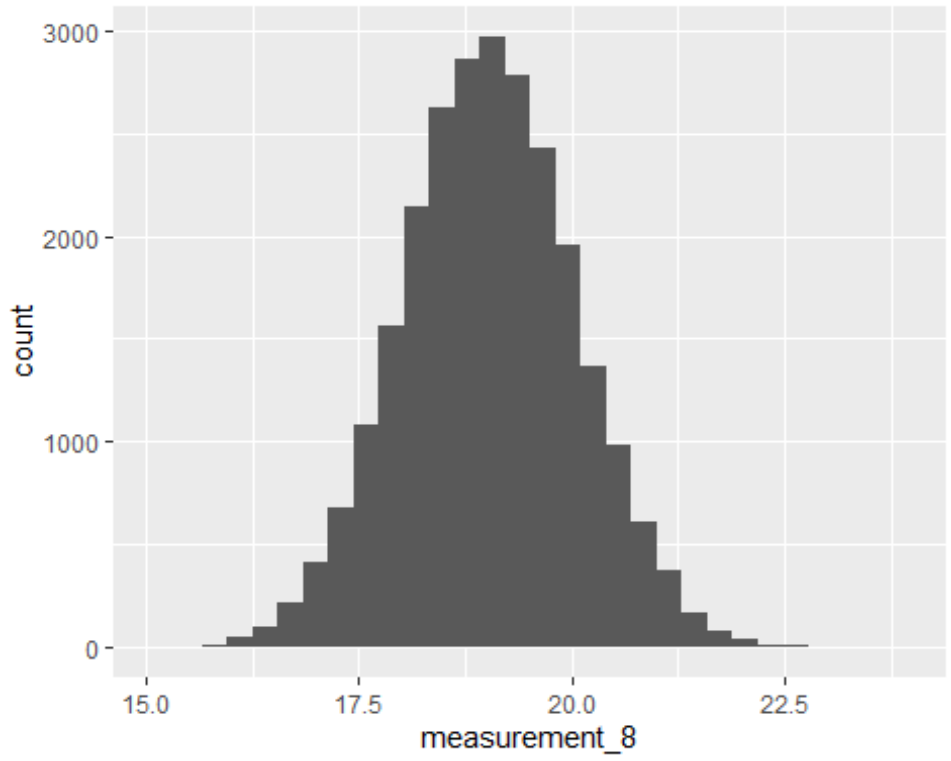
```
## Warning: Removed 937 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_8)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 1047 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



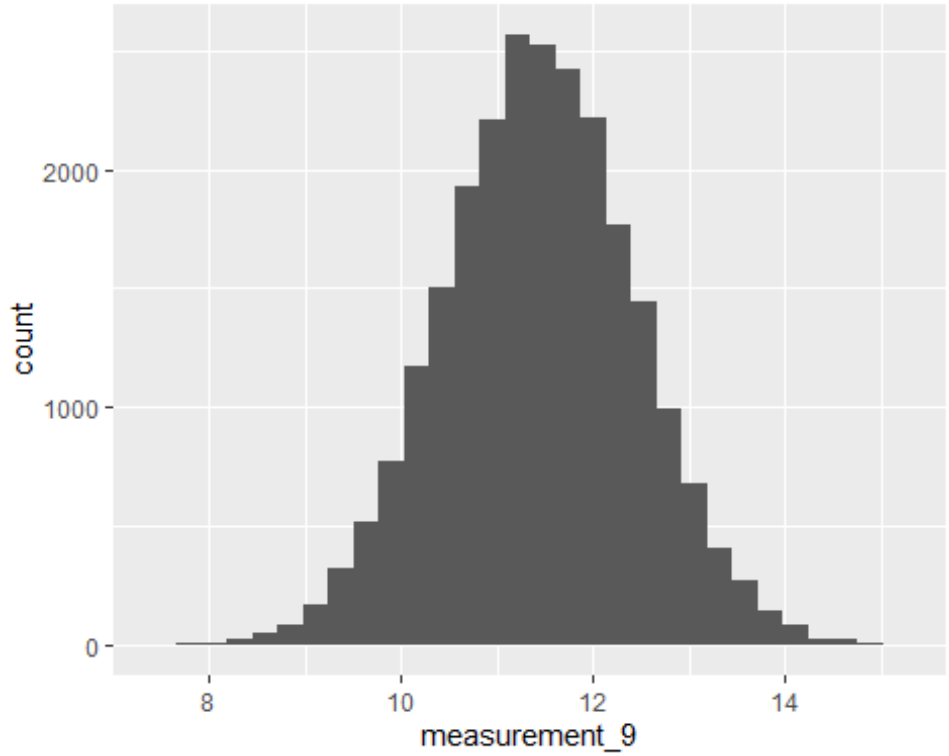
Remove outliers

```
train = train %>% filter (measurement_8 < 23.5)

ggplot(train, aes(x=measurement_9)) + geom_histogram()

## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.

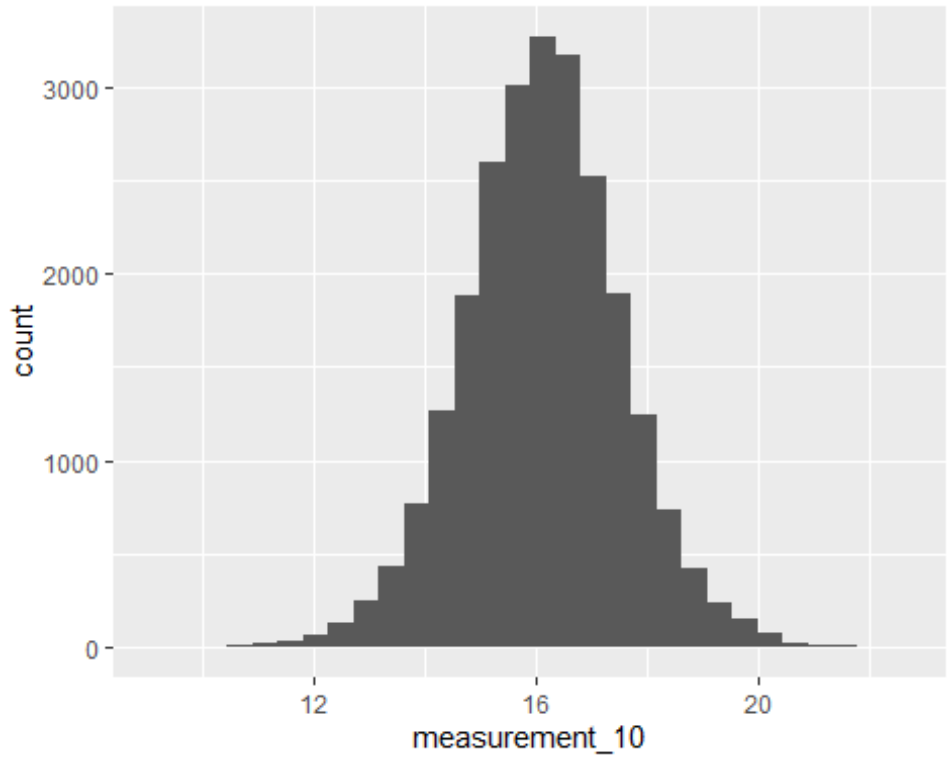
## Warning: Removed 1171 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_10)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

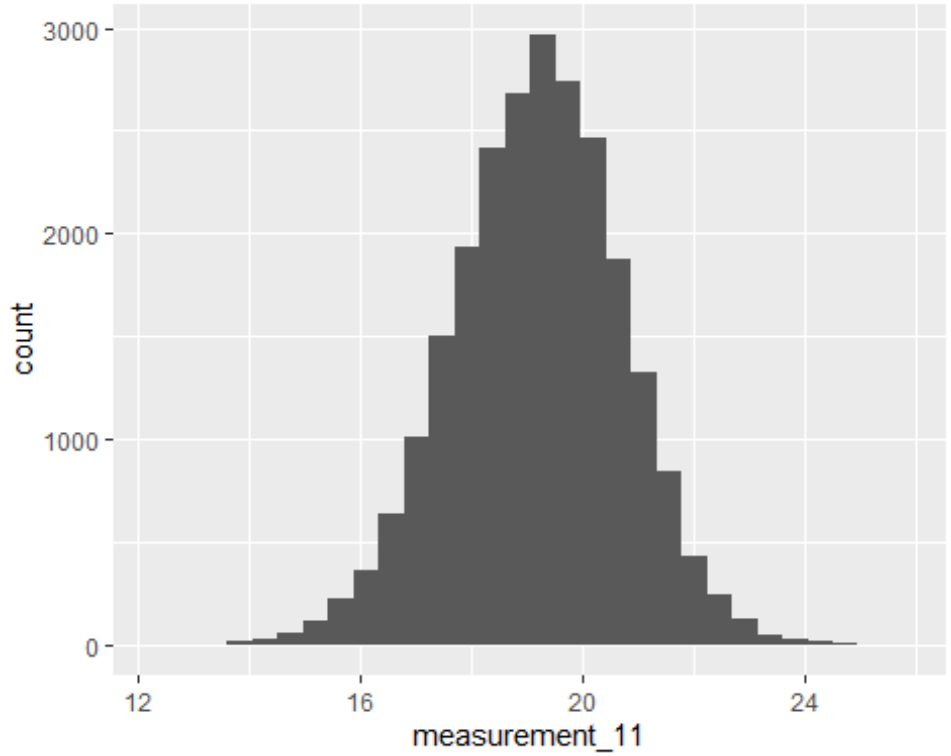
```
## Warning: Removed 1245 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_11)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

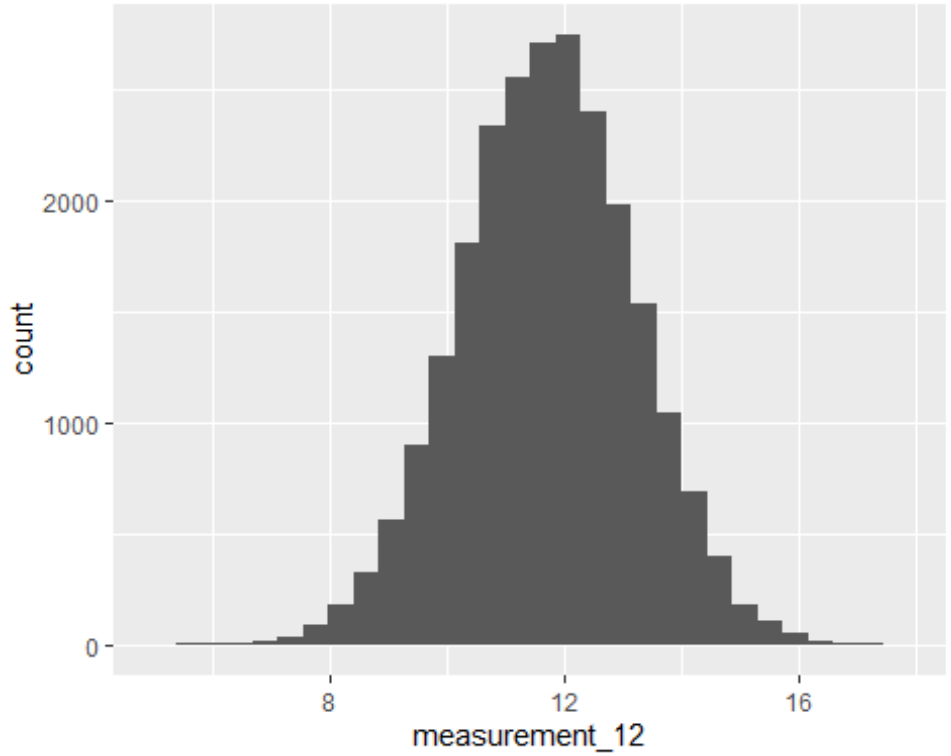
```
## Warning: Removed 1409 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_12)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

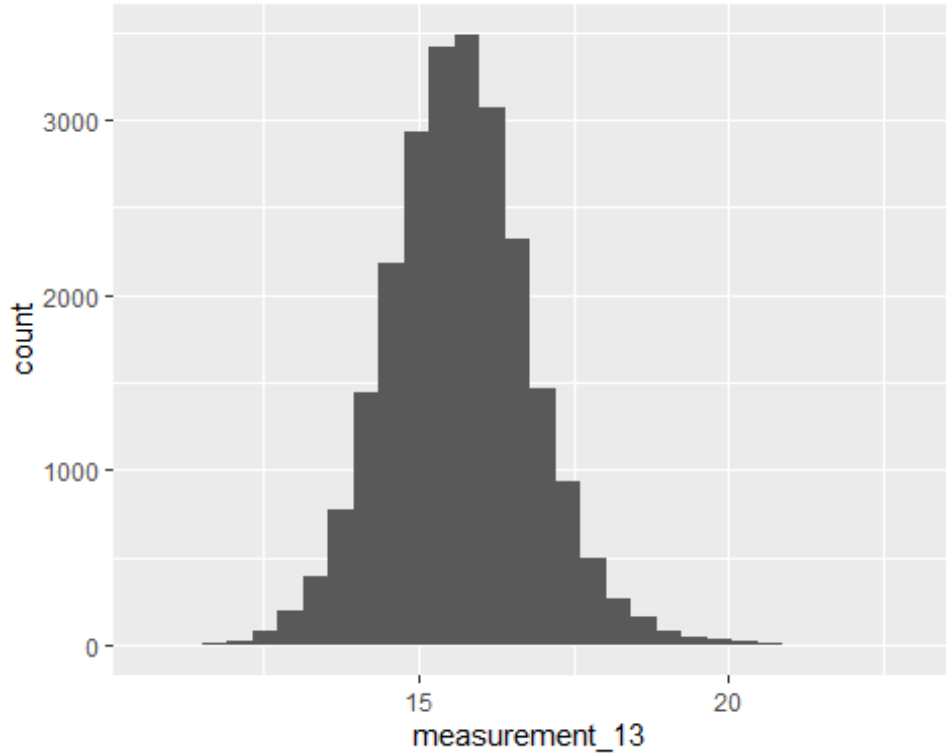
```
## Warning: Removed 1544 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_13)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

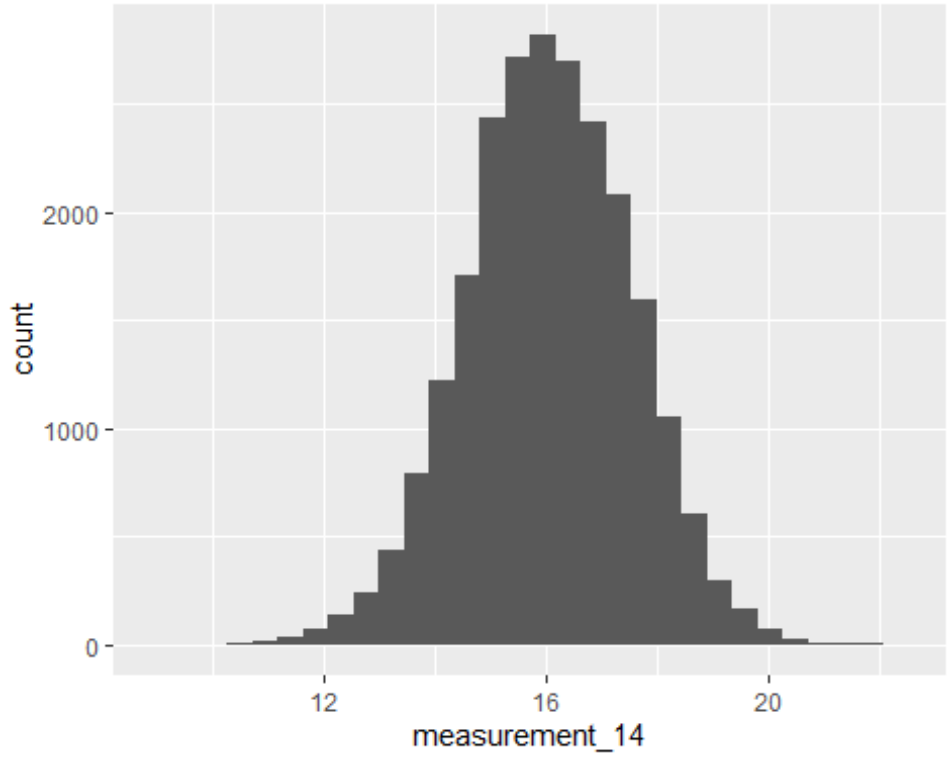
```
## Warning: Removed 1718 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_14)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 1802 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



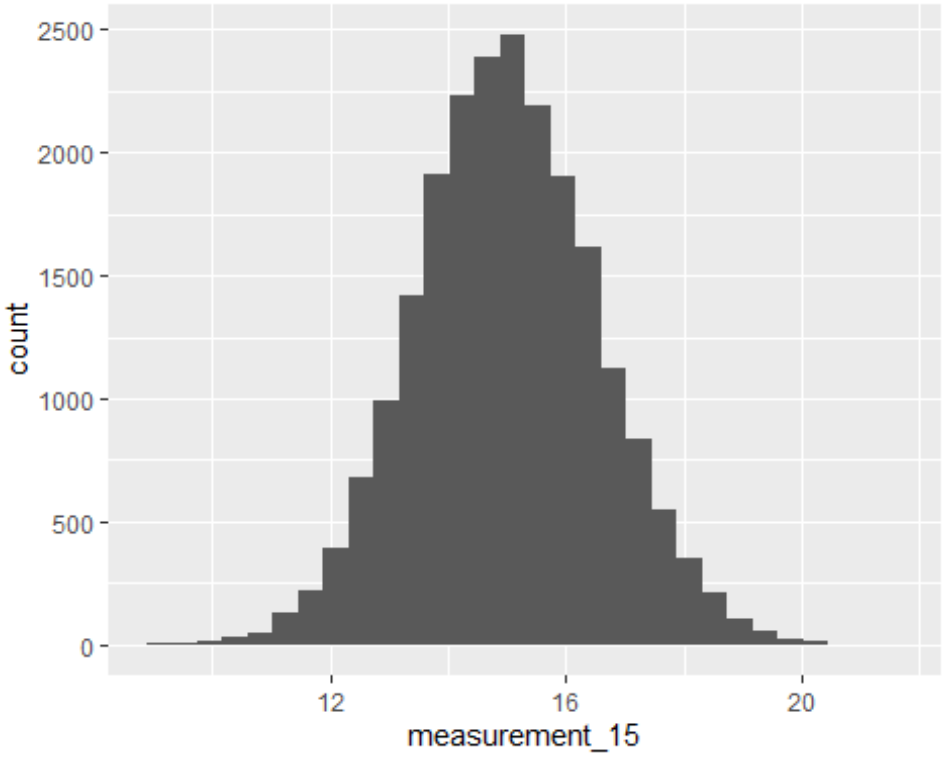
Remove outliers

```
train = train %>% filter(measurement_14 > 8)

ggplot(train, aes(x=measurement_15)) + geom_histogram()

## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.

## Warning: Removed 1781 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



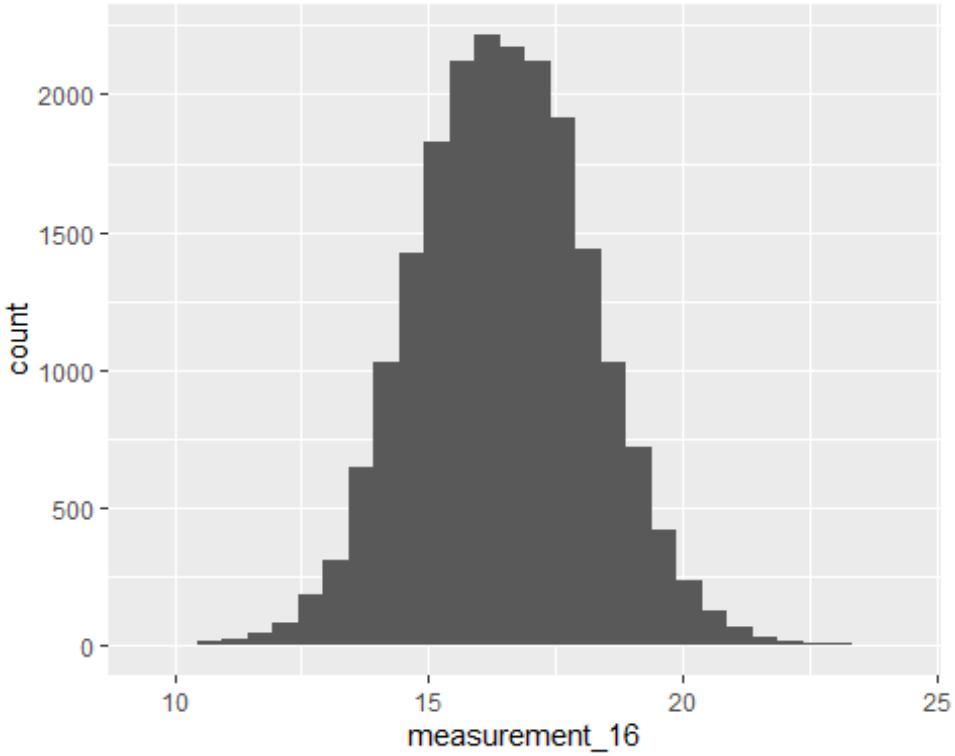
Remove outliers

```
train = train %>% filter(measurement_15 < 23)

ggplot(train, aes(x=measurement_16)) + geom_histogram()

## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.

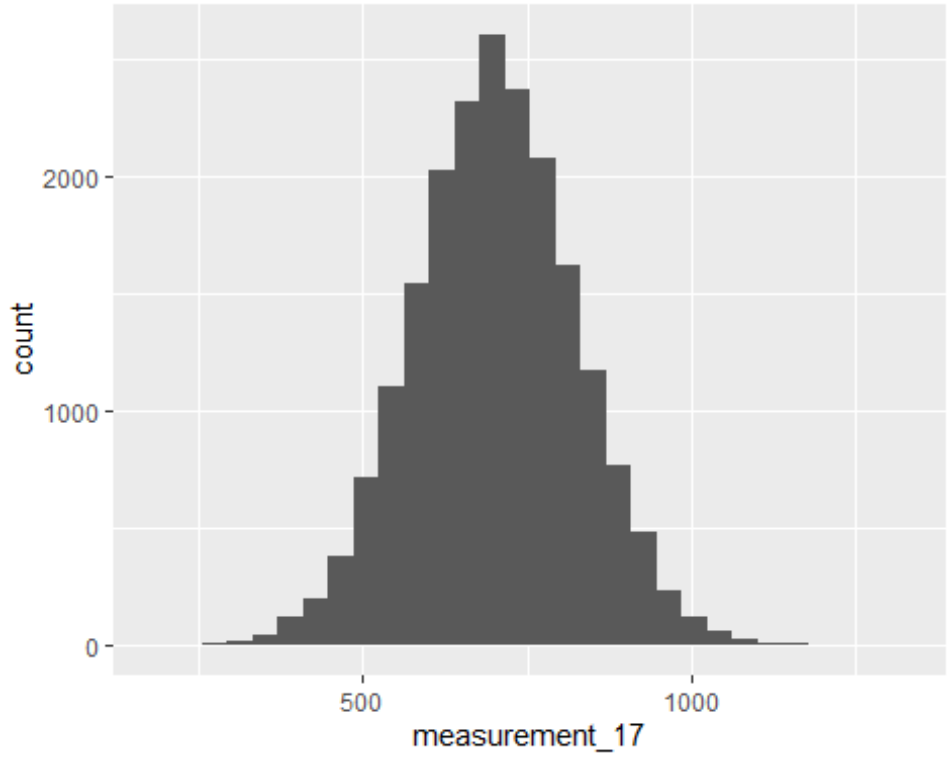
## Warning: Removed 1734 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
ggplot(train, aes(x=measurement_17)) + geom_histogram()
```

```
## `stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
## Warning: Removed 1891 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



Remove outliers

```
train = train %>% filter(measurement_17 < 1500)
```

Impute data into new dataframe

```
imp = mice(train, m=5, method = "pmm")

##
##  iter imp variable
##   1   1  loading  measurement_3*  measurement_4*  measurement_5*  measurem
ent_6*  measurement_7*  measurement_9*  measurement_10*  measurement_11*  mea
surement_12*  measurement_13*  measurement_16*
##   1   2  loading*  measurement_3*  measurement_4*  measurement_5*  measure
ment_6*  measurement_7*  measurement_9*  measurement_10*  measurement_11*  me
asurement_12*  measurement_13*  measurement_16*
##   1   3  loading*  measurement_3*  measurement_4*  measurement_5*  measure
ment_6*  measurement_7*  measurement_9*  measurement_10*  measurement_11*  me
asurement_12*  measurement_13*  measurement_16*
##   1   4  loading*  measurement_3  measurement_4*  measurement_5*  measurem
ent_6*  measurement_7*  measurement_9*  measurement_10*  measurement_11  meas
urement_12*  measurement_13*  measurement_16*
##   1   5  loading*  measurement_3*  measurement_4*  measurement_5*  measure
ment_6*  measurement_7*  measurement_9*  measurement_10*  measurement_11*  me
asurement_12*  measurement_13*  measurement_16*
##   2   1  loading*  measurement_3*  measurement_4*  measurement_5*  measure
ment_6*  measurement_7*  measurement_9*  measurement_10  measurement_11*  mea
```

```
surement_12*  measurement_13  measurement_16*
##   2   2 loading* measurement_3* measurement_4* measurement_5* measure
ment_6* measurement_7* measurement_9* measurement_10* measurement_11* me
asurement_12* measurement_13* measurement_16
##   2   3 loading* measurement_3* measurement_4* measurement_5* measure
ment_6* measurement_7* measurement_9* measurement_10* measurement_11* me
asurement_12* measurement_13* measurement_16
##   2   4 loading* measurement_3* measurement_4* measurement_5* measure
ment_6* measurement_7* measurement_9* measurement_10* measurement_11* me
asurement_12* measurement_13* measurement_16*
##   2   5 loading measurement_3* measurement_4* measurement_5* measurem
ent_6* measurement_7 measurement_9* measurement_10* measurement_11* meas
urement_12* measurement_13* measurement_16*
##   3   1 loading* measurement_3* measurement_4* measurement_5* measure
ment_6* measurement_7* measurement_9* measurement_10* measurement_11* me
asurement_12* measurement_13* measurement_16*
##   3   2 loading* measurement_3* measurement_4* measurement_5 measurem
ent_6* measurement_7* measurement_9* measurement_10* measurement_11* mea
surement_12* measurement_13* measurement_16
##   3   3 loading* measurement_3 measurement_4* measurement_5* measurem
ent_6 measurement_7* measurement_9* measurement_10* measurement_11* meas
urement_12* measurement_13* measurement_16*
##   3   4 loading* measurement_3* measurement_4* measurement_5* measure
ment_6* measurement_7 measurement_9* measurement_10* measurement_11* mea
surement_12* measurement_13* measurement_16*
##   3   5 loading* measurement_3* measurement_4* measurement_5* measure
ment_6* measurement_7* measurement_9* measurement_10* measurement_11 mea
surement_12* measurement_13* measurement_16
##   4   1 loading* measurement_3* measurement_4* measurement_5 measurem
ent_6* measurement_7 measurement_9* measurement_10* measurement_11 measu
rement_12* measurement_13* measurement_16*
##   4   2 loading* measurement_3 measurement_4* measurement_5* measurem
ent_6* measurement_7* measurement_9* measurement_10* measurement_11* mea
surement_12 measurement_13* measurement_16*
##   4   3 loading* measurement_3* measurement_4* measurement_5* measure
ment_6* measurement_7* measurement_9* measurement_10* measurement_11* me
asurement_12* measurement_13* measurement_16*
##   4   4 loading* measurement_3* measurement_4* measurement_5* measure
ment_6* measurement_7* measurement_9* measurement_10* measurement_11 mea
surement_12* measurement_13* measurement_16*
##   4   5 loading measurement_3* measurement_4* measurement_5* measurem
ent_6* measurement_7* measurement_9* measurement_10* measurement_11* mea
surement_12* measurement_13 measurement_16
##   5   1 loading* measurement_3* measurement_4* measurement_5* measure
ment_6* measurement_7* measurement_9* measurement_10* measurement_11 mea
surement_12* measurement_13* measurement_16*
##   5   2 loading measurement_3* measurement_4* measurement_5* measurem
ent_6* measurement_7* measurement_9* measurement_10* measurement_11* mea
surement_12* measurement_13* measurement_16*
##   5   3 loading* measurement_3* measurement_4* measurement_5* measure
```

```
ment_6*  measurement_7*  measurement_9*  measurement_10*  measurement_11*  me
asurement_12*  measurement_13*  measurement_16*
##   5   4  loading*  measurement_3*  measurement_4*  measurement_5*  measure
ment_6*  measurement_7*  measurement_9*  measurement_10*  measurement_11*  me
asurement_12*  measurement_13  measurement_16*
##   5   5  loading*  measurement_3*  measurement_4*  measurement_5*  measure
ment_6*  measurement_7*  measurement_9*  measurement_10*  measurement_11*  me
asurement_12*  measurement_13*  measurement_16

## Warning: Number of logged events: 571

imp_train = complete(imp)
```

Univariate logistic regressions

```
summary(glm(failure ~ attribute_2, data = imp_train, family = binomial))

##
## Call:
## glm(formula = failure ~ attribute_2, family = binomial, data = imp_train)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.40266    0.08112 -17.292   <2e-16 ***
## attribute_2  0.01285    0.01169   1.099    0.272
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20683  on 20041  degrees of freedom
## AIC: 20687
##
## Number of Fisher Scoring iterations: 4

summary(glm(failure ~ attribute_3, data = imp_train, family = binomial))

##
## Call:
## glm(formula = failure ~ attribute_3, family = binomial, data = imp_train)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept) -1.10362    0.08675 -12.721   <2e-16 ***
## attribute_3 -0.02938    0.01181  -2.488   0.0129 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20678  on 20041  degrees of freedom
## AIC: 20682
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_0, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_0, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.352099   0.035518 -38.067   <2e-16 ***
## measurement_0  0.004894   0.004150   1.179    0.238
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20683  on 20041  degrees of freedom
## AIC: 20687
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_1, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_1, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.265725   0.037831 -33.457   <2e-16 ***
## measurement_1 -0.006104   0.004130  -1.478    0.139
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20682  on 20041  degrees of freedom
## AIC: 20686
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_2, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_2, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.368864   0.037128 -36.869   <2e-16 ***
## measurement_2  0.008479   0.005212   1.627    0.104
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20682  on 20041  degrees of freedom
## AIC: 20686
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_3, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_3, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.445639   0.307680  -4.699 2.62e-06 ***
## measurement_3  0.007303   0.017254   0.423    0.672
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20684  on 20041  degrees of freedom
## AIC: 20688
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_4, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_4, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept)   -1.19445    0.20255  -5.897  3.7e-09 ***
## measurement_4 -0.01032    0.01718  -0.600    0.548
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20684  on 20041  degrees of freedom
## AIC: 20688
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_5, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_5, family = binomial, data = imp_trai
n)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.63119    0.27965  -5.833 5.45e-09 ***
## measurement_5  0.01848    0.01634   1.131    0.258
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20683  on 20041  degrees of freedom
## AIC: 20687
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_6, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_6, family = binomial, data = imp_trai
n)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.77732    0.26498  -6.707 1.98e-11 ***
## measurement_6  0.02651    0.01517   1.747   0.0806 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20681  on 20041  degrees of freedom
## AIC: 20685
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_7, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_7, family = binomial, data = imp_trai
n)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.46672    0.18861  -7.777 7.45e-15 ***
## measurement_7  0.01281    0.01592   0.805    0.421
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20684  on 20041  degrees of freedom
## AIC: 20688
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_8, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_8, family = binomial, data = imp_trai
n)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -2.08700    0.32800  -6.363 1.98e-10 ***
## measurement_8  0.04052    0.01720   2.356   0.0184 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20679  on 20041  degrees of freedom
## AIC: 20683
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_9, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_9, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.214184   0.197542  -6.146 7.92e-10 ***
## measurement_9 -0.008875   0.017219  -0.515    0.606
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20684  on 20041  degrees of freedom
## AIC: 20688
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_10, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_10, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.264252   0.182292  -6.935 4.05e-12 ***
## measurement_10 -0.003158   0.011153  -0.283    0.777
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20684  on 20041  degrees of freedom
## AIC: 20688
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_11, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_11, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
```

```
## (Intercept)    -1.148486   0.218767  -5.250 1.52e-07 ***
## measurement_11 -0.008717   0.011377  -0.766    0.444
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20684  on 20041  degrees of freedom
## AIC: 20688
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_12, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_12, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.302677   0.122864 -10.603   <2e-16 ***
## measurement_12 -0.001119   0.010507  -0.107    0.915
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20684  on 20041  degrees of freedom
## AIC: 20688
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_13, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_13, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.07053    0.19139  -5.594 2.22e-08 ***
## measurement_13 -0.01543    0.01201  -1.285    0.199
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20683  on 20041  degrees of freedom
## AIC: 20687
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_14, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_14, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.461007   0.188026  -7.770 7.83e-15 ***
## measurement_14  0.009052   0.011654   0.777    0.437
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20684  on 20041  degrees of freedom
## AIC: 20688
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_15, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_15, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.15583    0.16785  -6.886 5.73e-12 ***
## measurement_15 -0.01066    0.01114  -0.957    0.339
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20683  on 20041  degrees of freedom
## AIC: 20687
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_16, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_16, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.238645   0.149064  -1.601    0.109
## measurement_16 -0.065214   0.009005  -7.242 4.42e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20631  on 20041  degrees of freedom
## AIC: 20635
##
## Number of Fisher Scoring iterations: 4
```

```
summary(glm(failure ~ measurement_17, data = imp_train, family = binomial))
```

```
##
## Call:
## glm(formula = failure ~ measurement_17, family = binomial, data = imp_trai
n)
##
## Coefficients:
##                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -1.7219037  0.1008591 -17.072  < 2e-16 ***
## measurement_17  0.0005773  0.0001407   4.103 4.07e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20667  on 20041  degrees of freedom
## AIC: 20671
##
## Number of Fisher Scoring iterations: 4
```

Full logistic regression table

```
model <- glm(failure ~ ., data = imp_train, family = binomial)
summary(model)
```

```
##
## Call:
## glm(formula = failure ~ ., family = binomial, data = imp_train)
##
```

```
## Coefficients: (5 not defined because of singularities)
##                          Estimate Std. Error z value Pr(>|z|)
## (Intercept)           -5.482e-01  1.075e+00  -0.510  0.61000
## id                     5.408e-06  1.137e-05   0.476  0.63435
## product_codeB         -1.921e-01  8.674e-02  -2.215  0.02675 *
## product_codeC         -1.561e-01  1.359e-01  -1.149  0.25071
## product_codeD         -1.820e-01  1.944e-01  -0.936  0.34911
## product_codeE         -2.436e-01  2.511e-01  -0.970  0.33192
## loading                7.047e-03  4.228e-04  16.668  < 2e-16 ***
## attribute_0material_5         NA         NA      NA       NA
## attribute_1material_5         NA         NA      NA       NA
## attribute_1material_6         NA         NA      NA       NA
## attribute_2                   NA         NA      NA       NA
## attribute_3                   NA         NA      NA       NA
## measurement_0          8.831e-04  4.631e-03   0.191  0.84876
## measurement_1         -6.815e-04  4.808e-03  -0.142  0.88728
## measurement_2          1.298e-02  5.541e-03   2.343  0.01915 *
## measurement_3          2.563e-03  1.758e-02   0.146  0.88407
## measurement_4         -2.237e-02  1.811e-02  -1.235  0.21666
## measurement_5         -1.890e-02  1.992e-02  -0.949  0.34275
## measurement_6          2.708e-03  1.674e-02   0.162  0.87152
## measurement_7         -8.684e-03  1.799e-02  -0.483  0.62922
## measurement_8         -2.057e-03  2.257e-02  -0.091  0.92739
## measurement_9         -1.869e-02  1.779e-02  -1.051  0.29347
## measurement_10        -4.225e-03  1.139e-02  -0.371  0.71067
## measurement_11        -5.533e-03  1.177e-02  -0.470  0.63820
## measurement_12         2.820e-03  1.091e-02   0.258  0.79603
## measurement_13        -7.510e-03  1.226e-02  -0.613  0.54007
## measurement_14        -6.466e-04  1.188e-02  -0.054  0.95659
## measurement_15        -7.553e-04  1.164e-02  -0.065  0.94826
## measurement_16        -6.421e-02  9.256e-03  -6.937 4.01e-12 ***
## measurement_17         7.139e-04  2.358e-04   3.028  0.00246 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 20684  on 20042  degrees of freedom
## Residual deviance: 20319  on 20018  degrees of freedom
## AIC: 20369
##
## Number of Fisher Scoring iterations: 4

tidy(model) %>%
  arrange(p.value)

## # A tibble: 30 × 5
##    term            estimate std.error statistic  p.value
##    <chr>              <dbl>     <dbl>     <dbl>    <dbl>
##  1 loading          0.00705   0.000423    16.7   2.24e-62
```

```
##  2 measurement_16 -0.0642    0.00926     -6.94  4.01e-12
##  3 measurement_17  0.000714  0.000236     3.03  2.46e- 3
##  4 measurement_2   0.0130    0.00554      2.34  1.92e- 2
##  5 product_codeB  -0.192     0.0867      -2.22  2.67e- 2
##  6 measurement_4  -0.0224    0.0181      -1.24  2.17e- 1
##  7 product_codeC  -0.156     0.136       -1.15  2.51e- 1
##  8 measurement_9  -0.0187    0.0178      -1.05  2.93e- 1
##  9 product_codeE  -0.244     0.251       -0.970 3.32e- 1
## 10 measurement_5  -0.0189    0.0199      -0.949 3.43e- 1
## # ℹ 20 more rows
```